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Chapter #3:
Semiconductors

from **Microelectronic Circuits** Text
by Sedra and Smith
Oxford Publishing

Introduction

- **IN THIS CHAPTER WE WILL LEARN:**
  - The basic properties of semiconductors and, in particular, silicone – the material used to make most modern electronic circuits.
  - How doping a pure silicon crystal dramatically changes its electrical conductivity – the fundamental idea in underlying the use of semiconductors in the implementation of electronic devices.

Introduction

- **IN THIS CHAPTER WE WILL LEARN:**
  - The two mechanisms by which current flows in semiconductors – drift and diffusion charge carriers.
  - The structure and operation of the *pn* junction – a basic semiconductor structure that implements the diode and plays a dominant role in semiconductors.

3.1. Intrinsic Semiconductors

- **semiconductor** – a material whose conductivity lies between that of conductors (copper) and insulators (glass).
  - **single-element** – such as germanium and silicon.
  - **compound** – such as gallium-arsenide.

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3.1. Intrinsic Semiconductors

- **valence electron** – is an electron that participates in the formation of chemical bonds.
  - Atoms with one or two valence electrons more than a closed shell are highly reactive because the extra electrons are easily removed to form positive ions.
- **covalent bond** – is a form of chemical bond in which two atoms share a pair of atoms.
  - It is a stable balance of attractive and repulsive forces between atoms when they share electrons.

3.1. Intrinsic Semiconductors

- **silicon atom**
  - four valence electrons
  - requires four more to complete outermost shell
  - each pair of shared forms a covalent bond
  - the atoms form a lattice structure

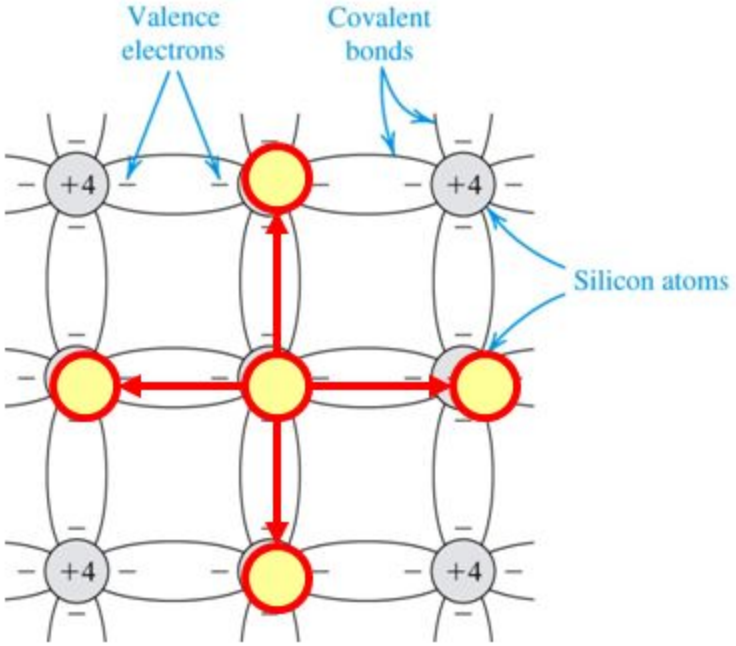


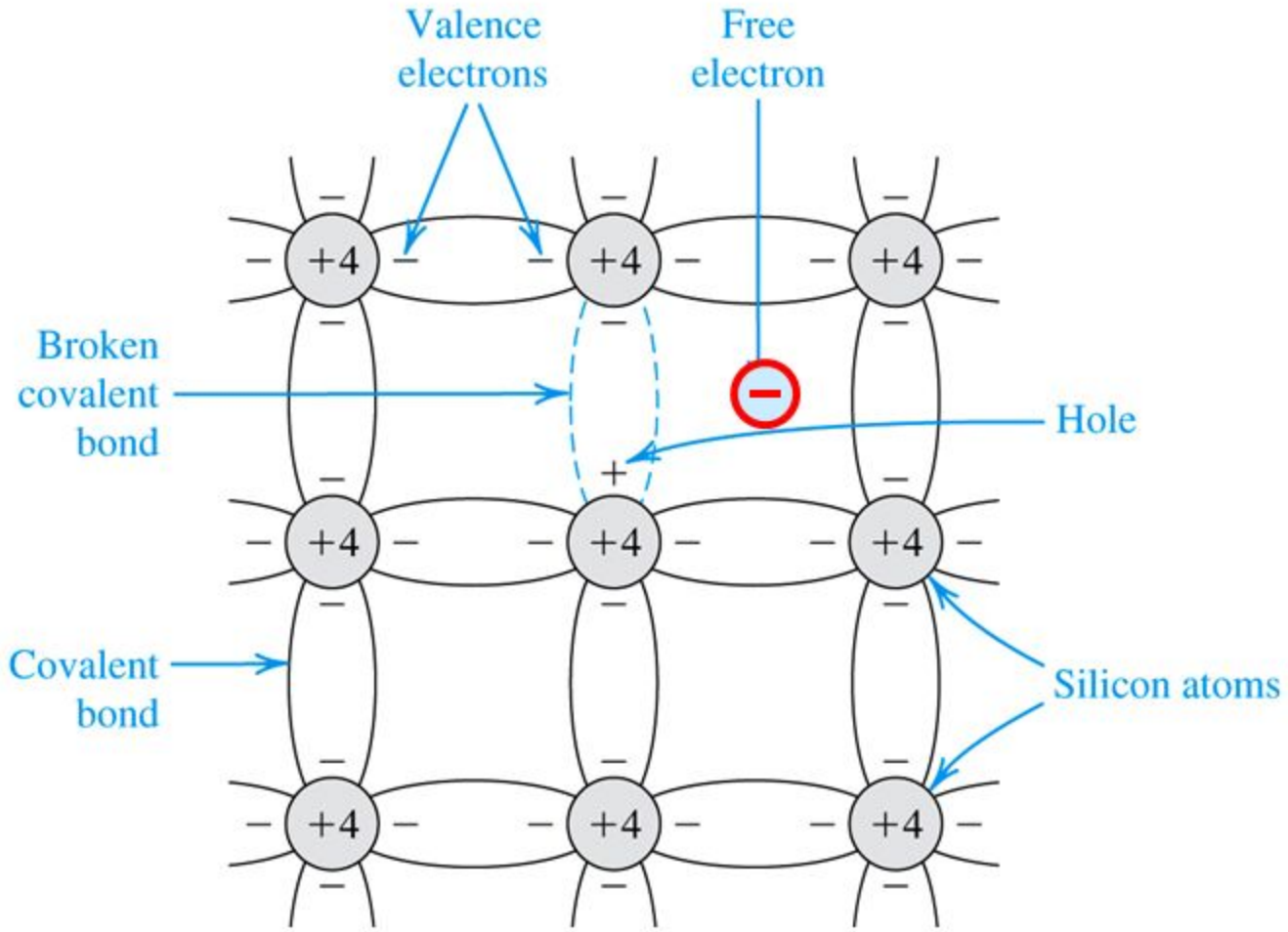
Figure 3.1 Two-dimensional representation of the silicon crystal. The circles represent the inner core of silicon atoms, with +4 indicating its positive charge of +4q, which is neutralized by the charge of the four valence electrons. Observe how the covalent bonds are formed by sharing of the valence electrons. At 0K, all bonds are intact and no free electrons are available for current conduction.

The process of freeing electrons, creating holes, and filling them **facilitates current flow**…

- silicon at low temps
  - all covalent bonds – are intact
  - no electrons – are available for conduction
  - conducitivity – is zero
- silicon at room temp
  - some covalent bonds – break, freeing an electron, creating hole, due to thermal energy
  - some electrons – will wander from their atoms, becoming available for conduction
  - conductivity – is greater than zero



Figure 3.2: At room temperature, some of the covalent bonds are broken by thermal generation. Each broken bond gives rise to a free electron and a hole, both of which become available for current conduction.



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3.1. **Intrinsic Semiconductors**

- **intrinsic semiconductor** – is one which is not doped
  - One example is pure silicon.
- **generation** – is the process of free electrons and holes being created.
  - **generation rate** – is speed with which this occurs.
- **recombination** – is the process of free electrons and holes disappearing.

Generation may be effected by **thermal energy**. As such, both generation and recombination rates will be (at least in part) a function of temperature.

3.1. Intrinsic Semiconductors

- **thermal generation** – effects a equal concentration of free electrons and holes.
  - Therefore, electrons move randomly throughout the material.
- In thermal equilibrium, generation and recombination rates are equal.

3.1. Intrinsic Semiconductors

- In thermal equilibrium, the behavior below applies…
  - n_i = number of free electrons and holes / unit volume
  - p = number of holes
  - n = number of free electrons

$$\text{(eq3.1)}\quad \underbrace{n_i = BT^{3/2}e^{-E_g/2kT}}_{\text{equal to } p \text{ and } n}$$

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3.1. **Intrinsic Semiconductors**

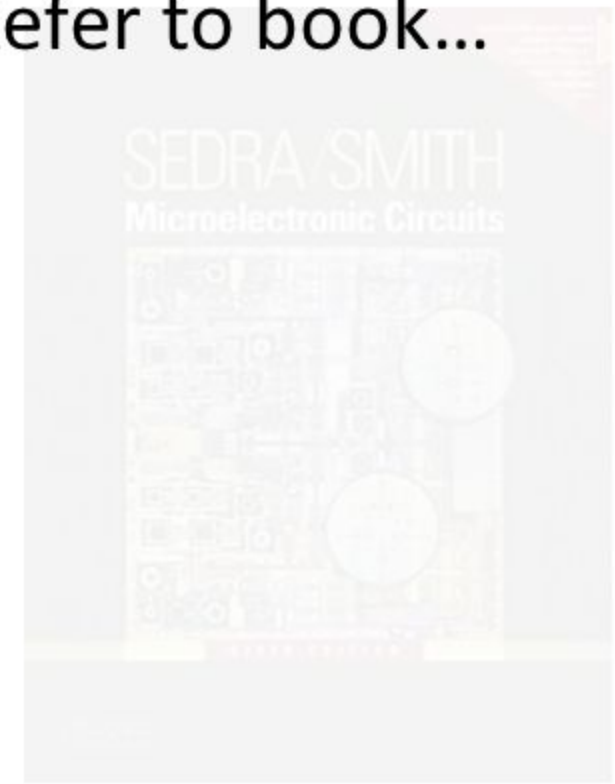
- n_i = number of free electrons and holes in a unit volume for intrinsic semiconductor
- B = parameter which is 7.3**E**15 $cm^{-3}K^{-3/2}$ for silicon
- T = temperature (K)
- E_g = bandgap energy which is 1.12eV for silicon
- k = Boltzman constant (8.62E-5 eV/K)

(eq3.1) $$\underbrace{n_i = BT^{3/2}e^{-E_g/2kT}}_{\text{equal to } p \text{ and } n}$$

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Example 3.1

- Refer to book…



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3.1. **Intrinsic Semiconductors**

- **Q:** Why can thermal generation not be used to effect meaningful current conduction?
  - **A:** Silicon crystal structure described previously is not sufficiently conductive at room temperature.
    - Additionally, a dependence on temperature is not desirable.

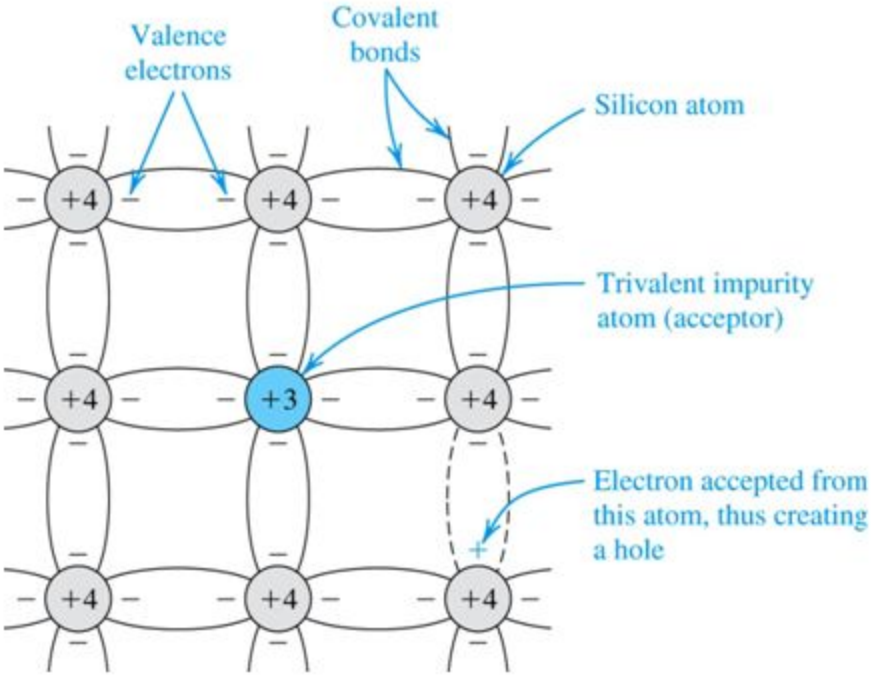
doping – is the intentional introduction of impurities into an extremely pure (intrinsic) semiconductor for the purpose changing carrier concentrations.

3.2. **Doped Semiconductors**

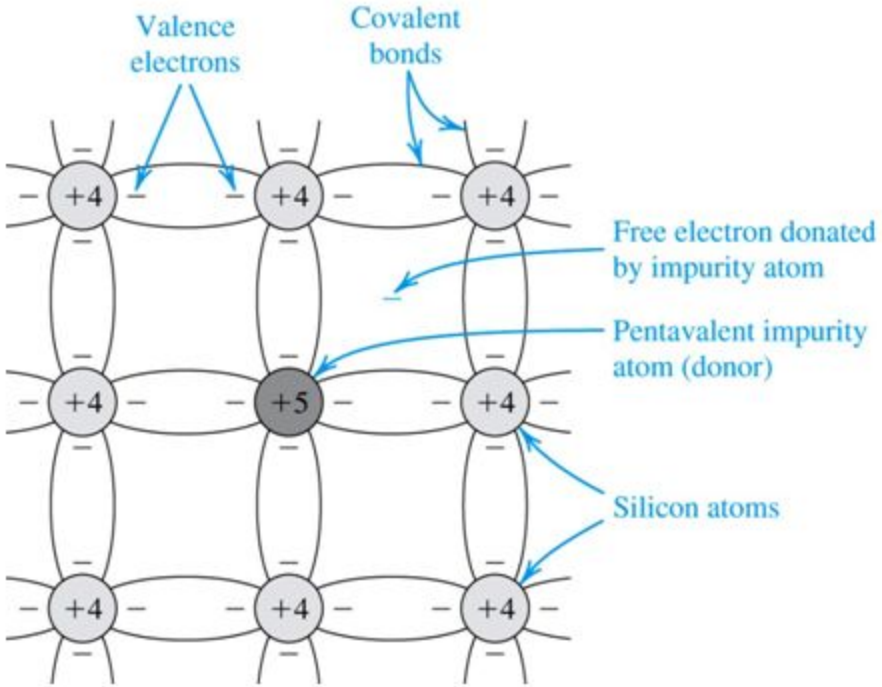
- ***p*-type semiconductor**
  - Silicon is doped with element having a valence of 3.
  - To increase the concentration of holes (p).
  - One example is boron, which is an acceptor.
- ***n*-type semiconductor**
  - Silicon is doped with element having a valence of 5.
  - To increase the concentration of free electrons (n).
  - One example is phosophorus, which is a donor.

3.2. **Doped Semiconductors**

- ***p*-type semiconductor**



- ***n*-type semiconductor**



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3.2. **Doped Semiconductors**

- ***p*-type doped semiconductor**
  - If N_A is much greater than n_i ...
    - concentration of acceptor atoms is N_A
  - Then the concentration of holes in the *p*-type is defined as below.

(eq3.6) $$\overbrace{(p_p) \approx (N_A)}^{\text{they will be equal...}}$$

(p_p): number holes in *p*-type

(N_A): number acceptor atoms

3.2. **Doped Semiconductors**

- ***n*-type doped semiconductor**
  - If N_D is much greater than n_i…
    - concentration of donor atoms is N_D
  - Then the concentration of electrons in the *n*-type is defined as below.

they will be equal…

The key here is that number of free electrons (aka. conductivity) is dependent on doping concentration, not temperature…

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3.2. **Doped Semiconductors**

- ***p*-type semiconductor**
  - **Q:** How can one find the concentration?
    - **A:** Use the formula to right, adapted for the *p*-type semiconductor.

action: combine this with equation on previous slide

$$\underbrace{p_p}_{\text{number of holes in } p\text{-type}} \times \underbrace{n_p}_{\text{number of free electrons in } p\text{-type}} = \underbrace{n_i^2}_{\text{number of free electrons and holes in thermal equil.}}$$

$$\text{(eq3.7)}\quad n_p \approx \frac{n_i^2}{n_A}$$

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3.2. **Doped Semiconductors**

- ***n*-type semiconductor**
  - **Q:** How can one find the concentration?
    - **A:** Use the formula to right, adapted for the *n*-type semiconductor.

action: combine this with equation on previous slide

$$\underbrace{p_n}_{\substack{\text{number of holes in n-type}}} \times \underbrace{n_n}_{\substack{\text{number of free electrons in n-type}}} = \underbrace{n_i^2}_{\substack{\text{number of free electrons and holes in thermal equil.}}}$$

$$\text{(eq3.5)}\ p_n \approx \frac{n_i^2}{n_D}$$

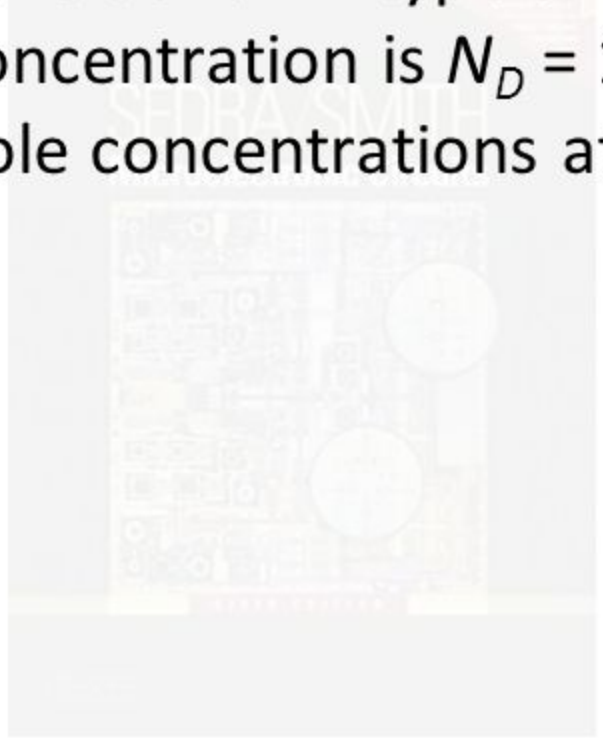
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3.2. **Doped Semiconductors**

- ***p*-type semiconductor**
  - n_p will have the same dependence on temperature as n_i^2
  - the concentration of holes (p_n) will be much larger than holes
  - holes are the majority charge carriers
  - free electrons are the minority charge carrier
- ***n*-type semiconductor**
  - p_n will have the same dependence on temperature as n_i^2
  - the concentration of free electrons (n_n) will be much larger than holes
  - electrons are the majority charge carriers
  - holes are the minority charge carrier

Example 3.2: Doped Semiconductor

- Consider an n-type silicon for which the dopant concentration is $N_D = 10^{17}/cm^3$. Find the electron and hole concentrations at $T = 300K$.



3.3.1. **Drift Current**

- **Q:** What happens when an electrical field (E) is applied to a semiconductor crystal?
  - **A:** Holes are accelerated in the direction of E, free electrons are repelled.
- **Q:** How is the velocity of these holes defined?

μ_p=hole mobility
E=electric field

$$(\text{eq3.8})\ v_{p-drift} = \mu_p E$$

μ_n=electron mobility
E=electric field

$$(\text{eq3.9})\ v_{n-drift} = -\mu_n E$$

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3.3.1 Drift Current

note that electrons move with velocity **2.5 times higher** than holes

E (volts / cm)

μ_p (cm^2/Vs) = 480 for silicon

μ_n (cm^2/Vs) = 1350 for silicon

μ_p=hole mobility
E=electric field

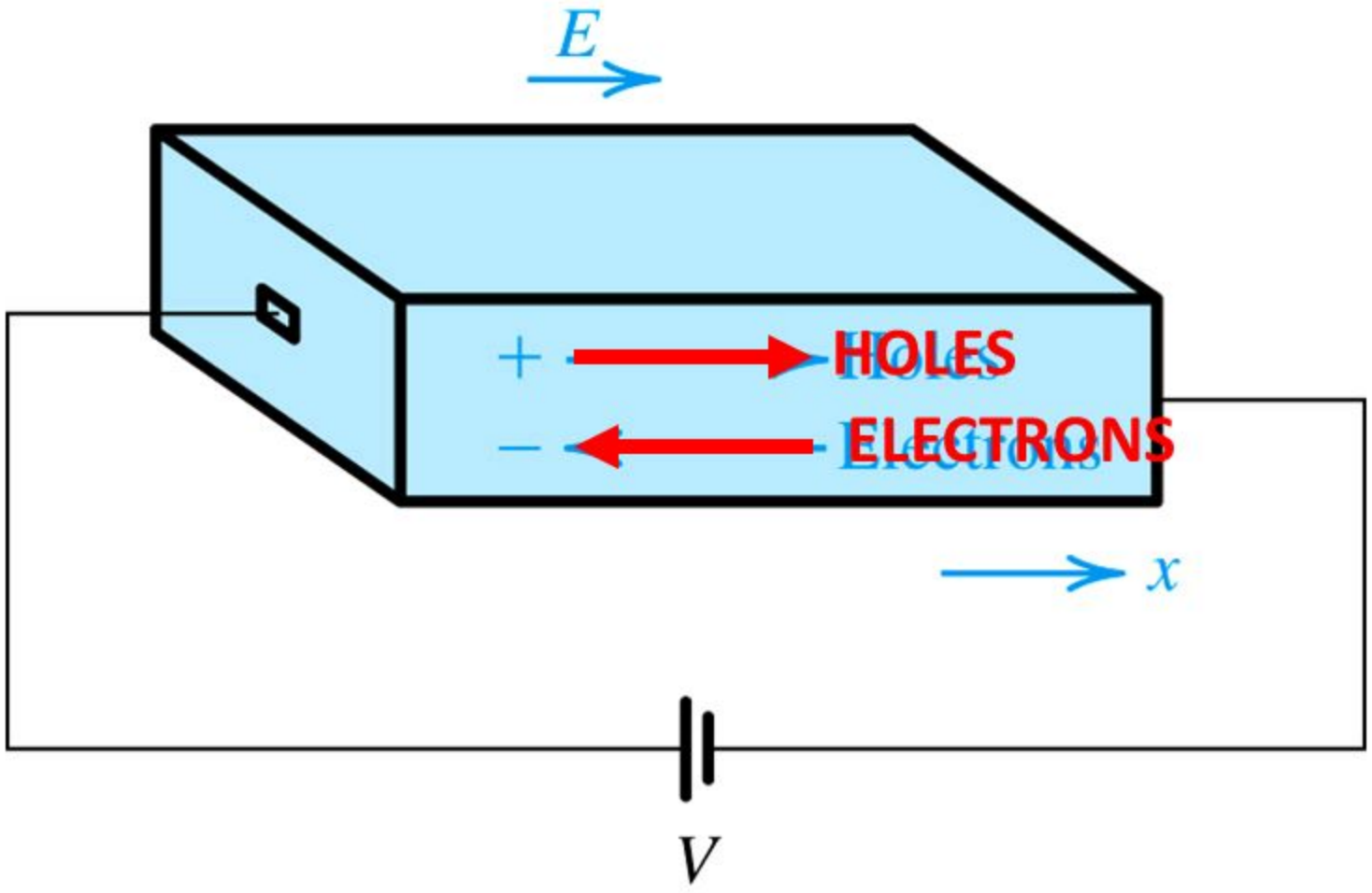
(eq3.8) $v_{p-drift} = \mu_p E$

μ_n=electron mobility
E=electric field

(eq3.9) $v_{n-drift} = -\mu_n E$

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Microelectronic Circuits by Adel S. Sedra and Kenneth C. Smith [0195323033]

Figure 3.5: An electric field *E* established in a bar of silicon causes the holes to drift in the direction of *E* and the free electrons to drift in the opposite direction. Both the hole and electron drift currents are in the direction of *E*.



3.3.1. **Drift Current**

- Assume that, for the single-crystal silicon bar on previous slide, the concentration of holes is defined as *p* and electrons as *n*.
- **Q:** What is the current component attributed to the flow of holes (not electrons)?

3.3.1. **Drift Current**

PART A: What is the current component attributed to the flow of holes (not electrons)?

- **step #1:** Consider a plane perpendicular to the *x* direction.
- **step #2:** Define the hole charge that crosses this plane.

I_p = current flow attributed to holes
A = cross-sectional area of silicon
q = magnitude of the electron charge
p = concentration of holes
$v_{p-drift}$ = drift velocity of holes

$$(\text{eq3.10})\ I_p = Aqpv_{p-drift}$$

3.3.1. **Drift Current**

- **step #3:** Substitute in $\mu_p E$.
- **step #4:** Define current density as $J_p = I_p / A$.

PART A: What is the current component attributed to the flow of holes (not electrons)?

I_p = current flow attributed to holes
A = cross-sectional area of silicon
q = magnitude of the electron charge
p = concentration of holes
μ_p = hole mobility
E = electric field

$$I_p = Aqp\mu_p E$$

$$(\text{eq3.11})\ J_p = qp\mu_p E$$

solution

3.3.1. **Drift Current**

- **Q:** What is the current component attributed to the flow of electrons (not holes)?
  - **A:** to the right…
- **Q:** How is total drift current defined?
  - **A:** to the right…

I_n = current flow attributed to electrons
A = cross-sectional area of silicon
q = magnitude of the electron charge
n = concentration of free electrons
μ_n = electron mobility
E = electric field

$$I_n = -Aqv_{n-drift}$$

$$(eq3.12)\ J_n = qn\mu_n E$$

$$(eq3.13)\ J = J_p + J_n = \underbrace{q(p\mu_p + n\mu_n)}_{\text{this is conductivity } (\sigma)} E$$

3.3.1. **Drift Current**

- **conductivity** (σ) – relates current density (J) and electrical field (E)
- **resistivity** (ρ) – relates current density (J) and electrical field (E)

Ohm's Law

(eq3.14) $J = \sigma E$

(eq3.16) $\sigma = q(p\mu_p + n\mu_n)$

(eq3.15) $J = E / \rho$

(eq3.17) $\rho = \dfrac{1}{q(p\mu_p + n\mu_n)}$

Example 3.3: Doped Semiconductors

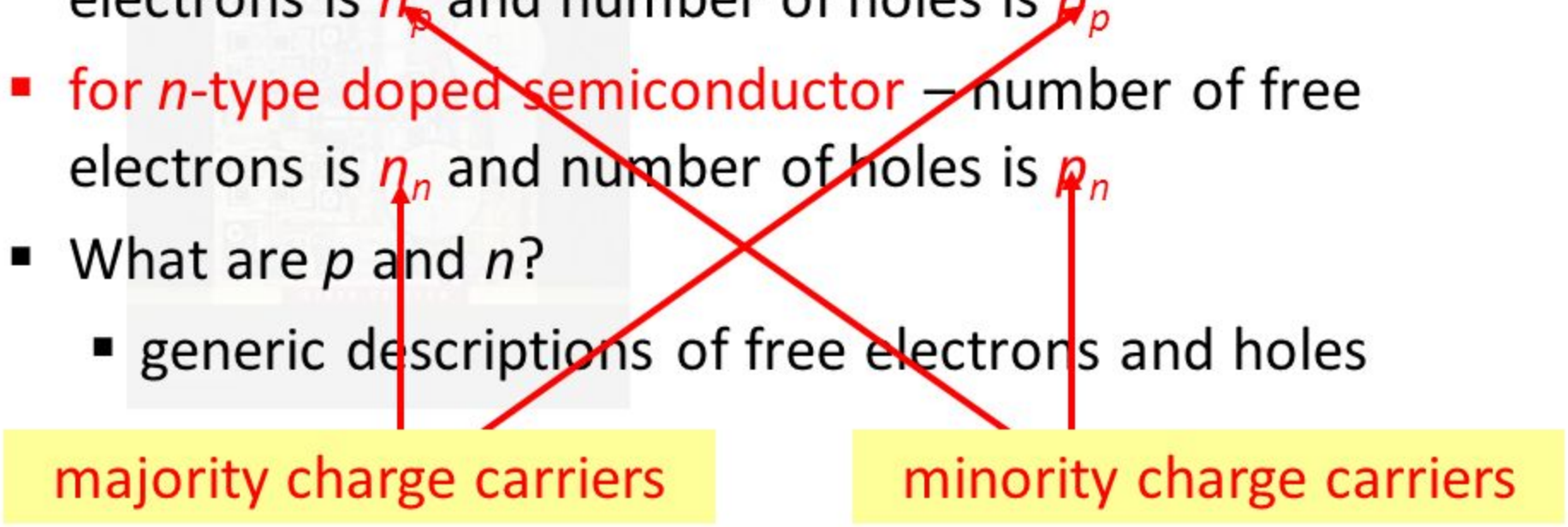
- **Q(a):** Find the resistivity of intrinsic silicon using following values – $\mu_n = 1350 cm^2/Vs$, $\mu_p = 480 cm^2/Vs$, $n_i = 1.5\mathbf{E}10/cm^3$.
- **Q(b):** Find the resistivity of *p*-type silicon with $N_A = 10^{16}/cm^2$ and using the following values – $\mu_n = 1110 cm^2/Vs$, $\mu_p = 400 cm^2/Vs$, $n_i = 1.5\mathbf{E}10/cm^3$

note that doping reduces carrier mobility

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Note…

- for intrinsic semiconductor – number of free electrons is n_i and number of holes is p_i
- for *p*-type doped semiconductor – number of free electrons is n_p and number of holes is p_p
- for *n*-type doped semiconductor – number of free electrons is n_n and number of holes is p_n
- What are *p* and *n*?
  - generic descriptions of free electrons and holes



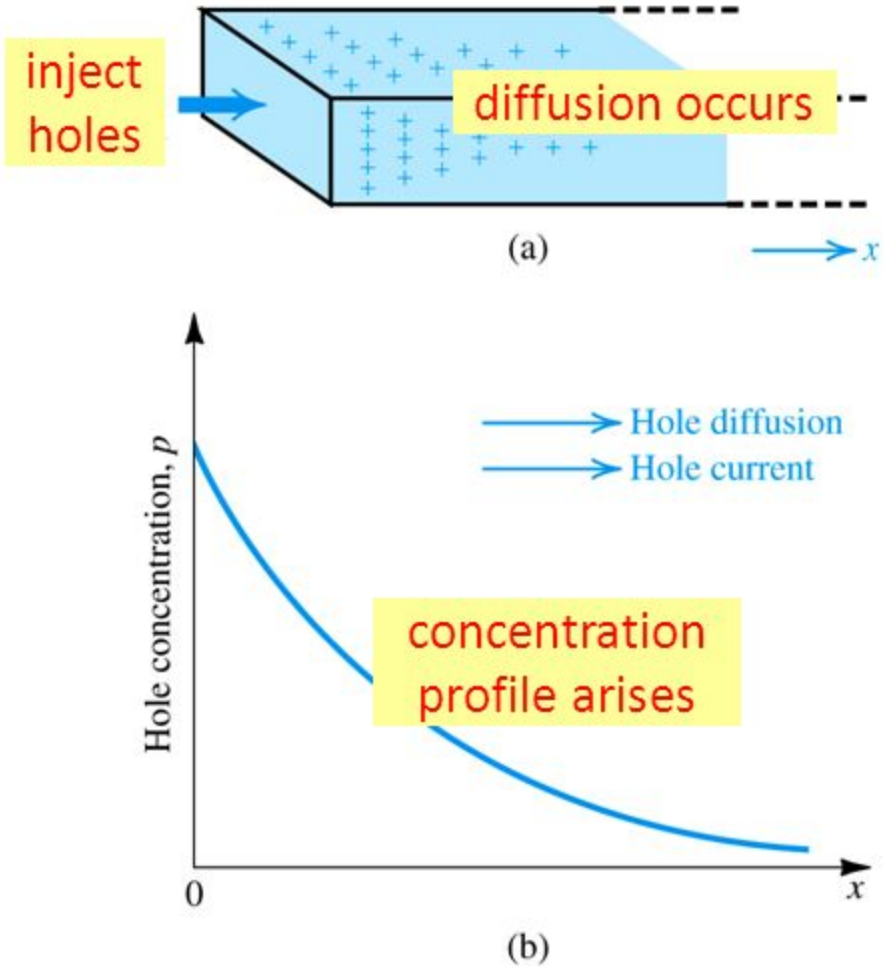
3.3.2. Diffusion Current

- **carrier diffusion** – is the flow of charge carriers from area of high concentration to low concentration.
  - It requires non-uniform distribution of carriers.
- **diffusion current** – is the current flow that results from diffusion.

3.3.2. **Diffusion Current**

Figure 3.6: A bar of silicon **(a)** into which holes are injected, thus creating the hole concentration profile along the *x* axis, shown in **(b)**. The holes diffuse in the positive direction of *x* and give rise to a hole-diffusion current in the same direction. Note that we are not showing the circuit to which the silicon bar is connected.

- Take the following example…
  - inject holes – By some unspecified process, one injects holes in to the left side of a silicon bar.
  - concentration profile arises – Because of this continuous hole inject, a concentration profile arises.
  - diffusion occurs – Because of this concentration gradient, holes will flow from left to right.



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3.3.2. Diffusion Current

- **Q:** How is diffusion current defined?

J_p = current flow density attributed to holes
q = magnitude of the electron charge
D_p = diffusion constant of holes ($12 cm^2/s$ for silicon)
$\mathbf{p}(x)$ = hole concentration at point x
$d\mathbf{p} / dx$ = gradient of hole concentration

(eq3.19) **hole diffusion current density :** $$J_p = -qD_p \frac{d\mathbf{p}(x)}{dx}$$

(eq3.20) **electron diffusion current density :** $$J_n = -qD_n \frac{d\mathbf{n}(x)}{dx}$$

J_n = current flow density attributed to free electrons
D_n = diffusion constant of electrons ($35 cm^2/s$ for silicon)
$\mathbf{n}(x)$ = free electron concentration at point x
$d\mathbf{n} / dx$ = gradient of free electron concentration

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Example 3.4: Diffusion

- Consider a bar of silicon in which a hole concentration **p**(x) described below is established.
- **Q(a):** Find the hole-current density J_p at $x = 0$.
- **Q(b):** Find current I_p.
  - Note the following parameters: $p_0 = 10^{16}/cm^3$, $L_p = 1\,\mu m$, $A = 100\,\mu m^2$

$$\mathbf{p}(x) = p_0 e^{-x/L_p}$$

3.3.3. Relationship Between D and μ?

- **Q:** What is the relationship between diffusion constant (D) and mobility (μ)?
  - **A: thermal voltage** (V_T)
- **Q:** What is this value?
  - **A:** at $T = 300K$, $V_T = 25.9mV$

the relationship between diffusion constant and mobility is defined by thermal voltage

(eq3.21) $$\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p} = V_T$$

known as Einstein Relationship

- drift current density (J_{drift})
  - effected by – an electric field (E).
- diffusion current density (J_{diff})
  - effected by – concentration gradient in free electrons and holes.

A= cross-sectional area of silicon, q= magnitude of the electron charge,
p= concentration of holes, n= concentration of free electrons,
μ_p= hole mobility, μ_n= electron mobility, E= electric field

drift current density : $J_{drift} = J_{p-drift} + J_{n-drift} = q(p\mu_p + n\mu_n)E$

diffusion current density : $J_{diff} = J_{p-diff} + J_{n-diff} = -qD_p \frac{d\mathbf{p}(x)}{dx} - qD_n \frac{d\mathbf{n}(x)}{dx}$

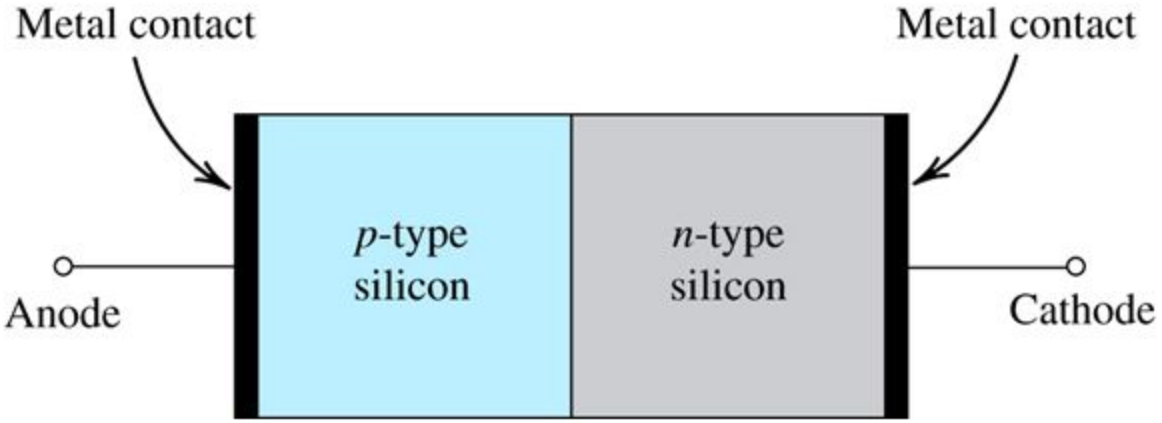
D_p= diffusion constant of holes (12cm^2/s for silicon), D_n= diffusion constant of electrons (35cm^2/s for silicon),
$\mathbf{p}(x)$= hole concentration at point x, $\mathbf{n}(x)$= free electron concentration at point x,
$d\mathbf{p}/dx$= gradient of hole concentration, $d\mathbf{n}/dx$= gradient of free electron concentration

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3.4.1. **Physical Structure**

- ***pn* junction** structure
  - *p*-type semiconductor
  - *n*-type semiconductor
  - metal contact for connection

Figure 3.8: Simplified physical structure of the *pn* junction. (Actual geometries are given in Appendix A.) As the *pn* junction implements the junction diode, its terminals are labeled anode and cathode.



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3.4.2. **Operation with Open-Circuit Terminals**

- **Q:** What is state of *pn* junction with open-circuit terminals?
- **A:** Read the below…
  - *p*-type material contains majority of holes
    - these holes are neutralized by equal amount of bound negative charge
  - *n*-type material contains majority of free electrons
    - these electrons are neutralized by equal amount of bound positive charge

3.4.2. Operation with Open-Circuit Terminals

- **bound charge**
  - charge of opposite polarity to free electrons / holes of a given material
  - neutralizes the electrical charge of these majority carriers
  - does not affect concentration gradients

3.4.2. **Operation with Open-Circuit Terminals**

- **Q:** What happens when a *pn*-junction is newly formed – aka. when the *p*-type and *n*-type semiconductors first touch one another?
  - **A:** See following slides…

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Step #1: The *p*-type and *n*-type semiconductors are joined at the junction.

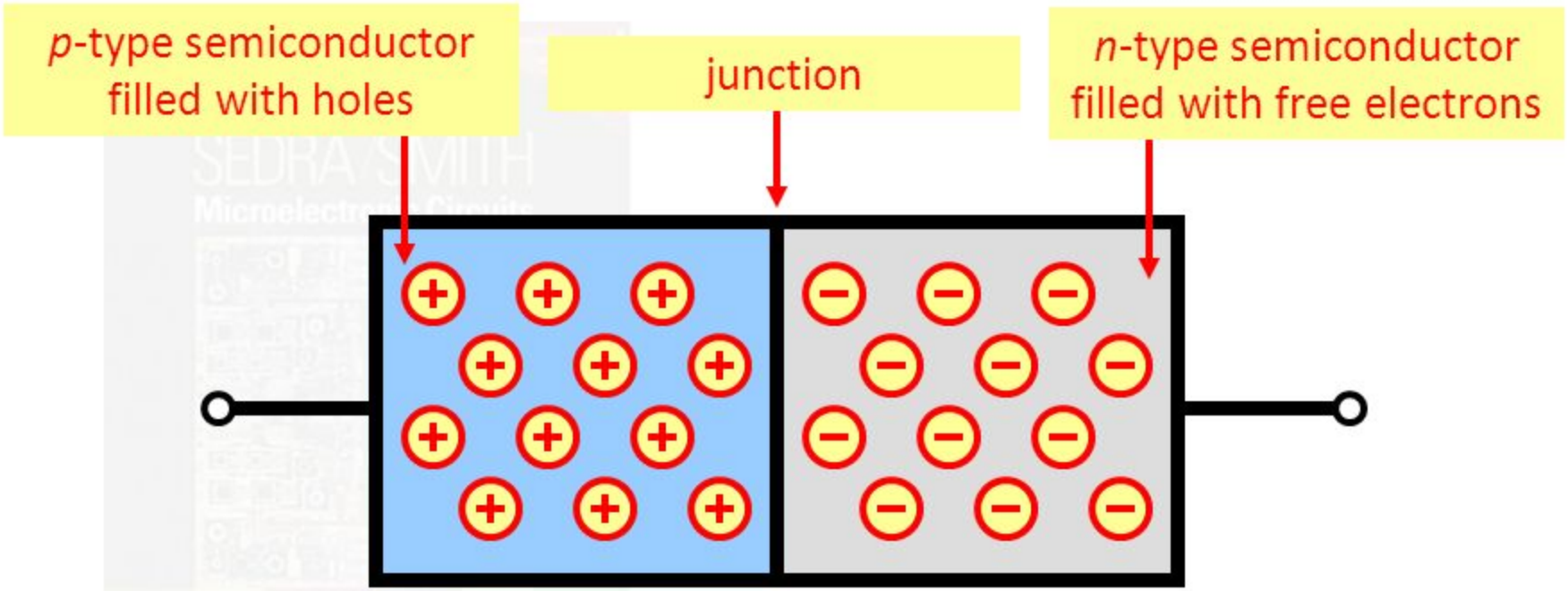


Figure: The *pn* junction with no applied voltage (open-circuited terminals).

Step #1A: Bound charges are attracted (from environment) by free electrons and holes in the *p*-type and *n*-type semiconductors, respectively. They remain weakly "bound" to these majority carriers; however, they **do not recombine**.

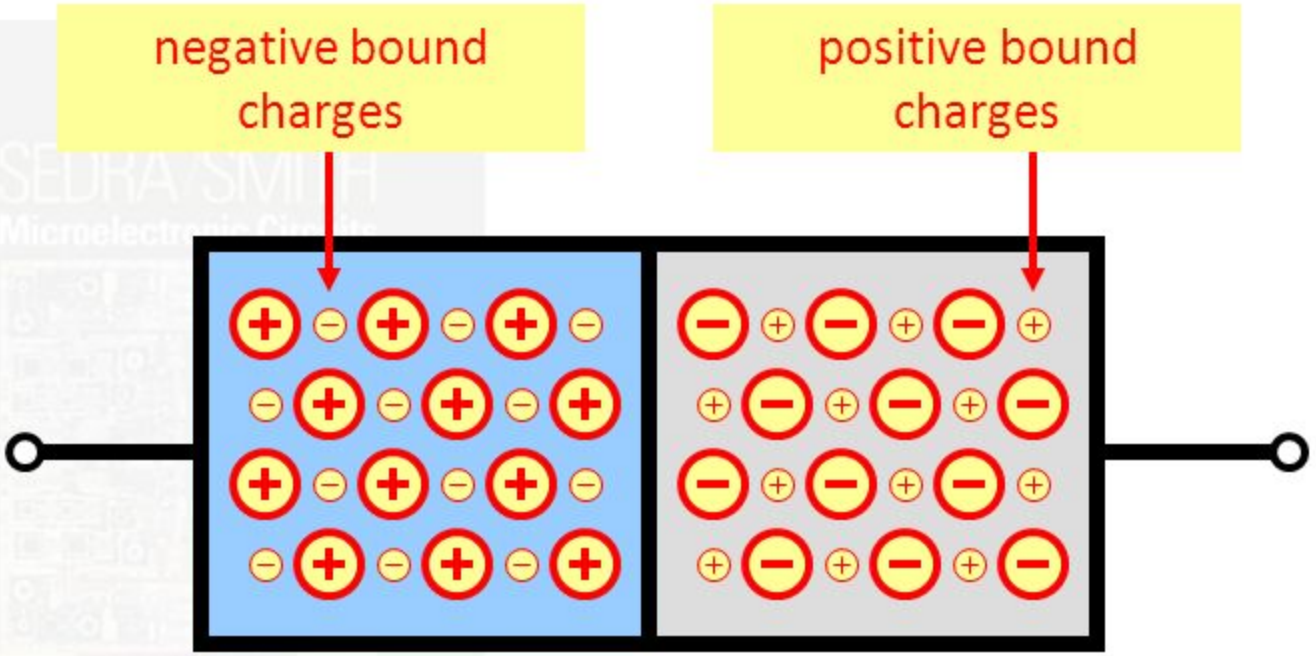


Figure: The *pn* junction with no applied voltage (open-circuited terminals).

Step #2: Diffusion begins. Those free electrons and holes which are closest to the junction will recombine and, essentially, eliminate one another.

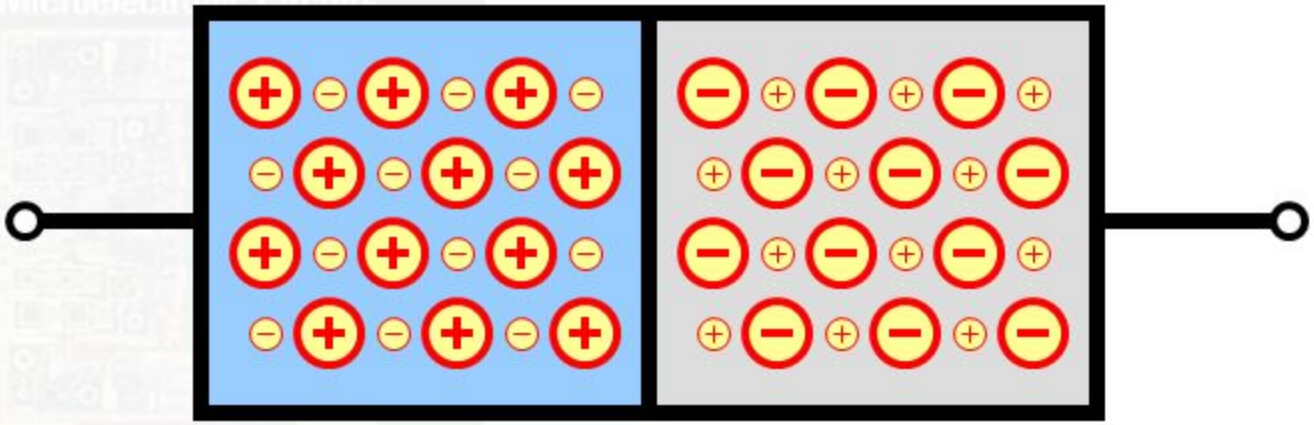


Figure: The *pn* junction with no applied voltage (open-circuited terminals).

Step #3: The **depletion region** begins to form – as diffusion occurs and free electrons recombine with holes.

The depletion region is filled with "uncovered" bound charges – who have lost the majority carriers to which they were linked.

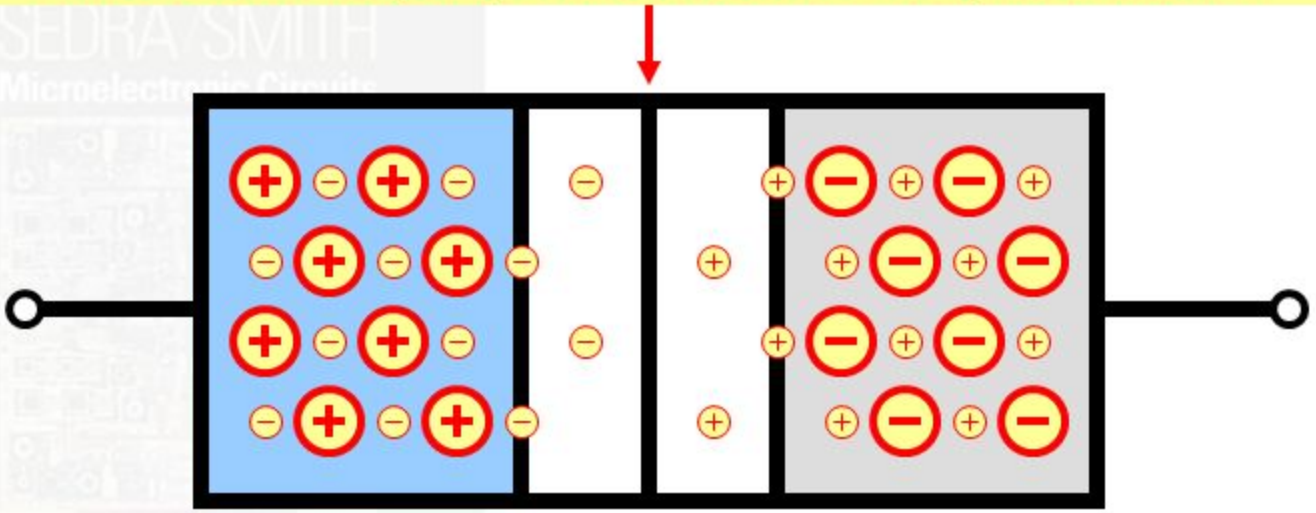
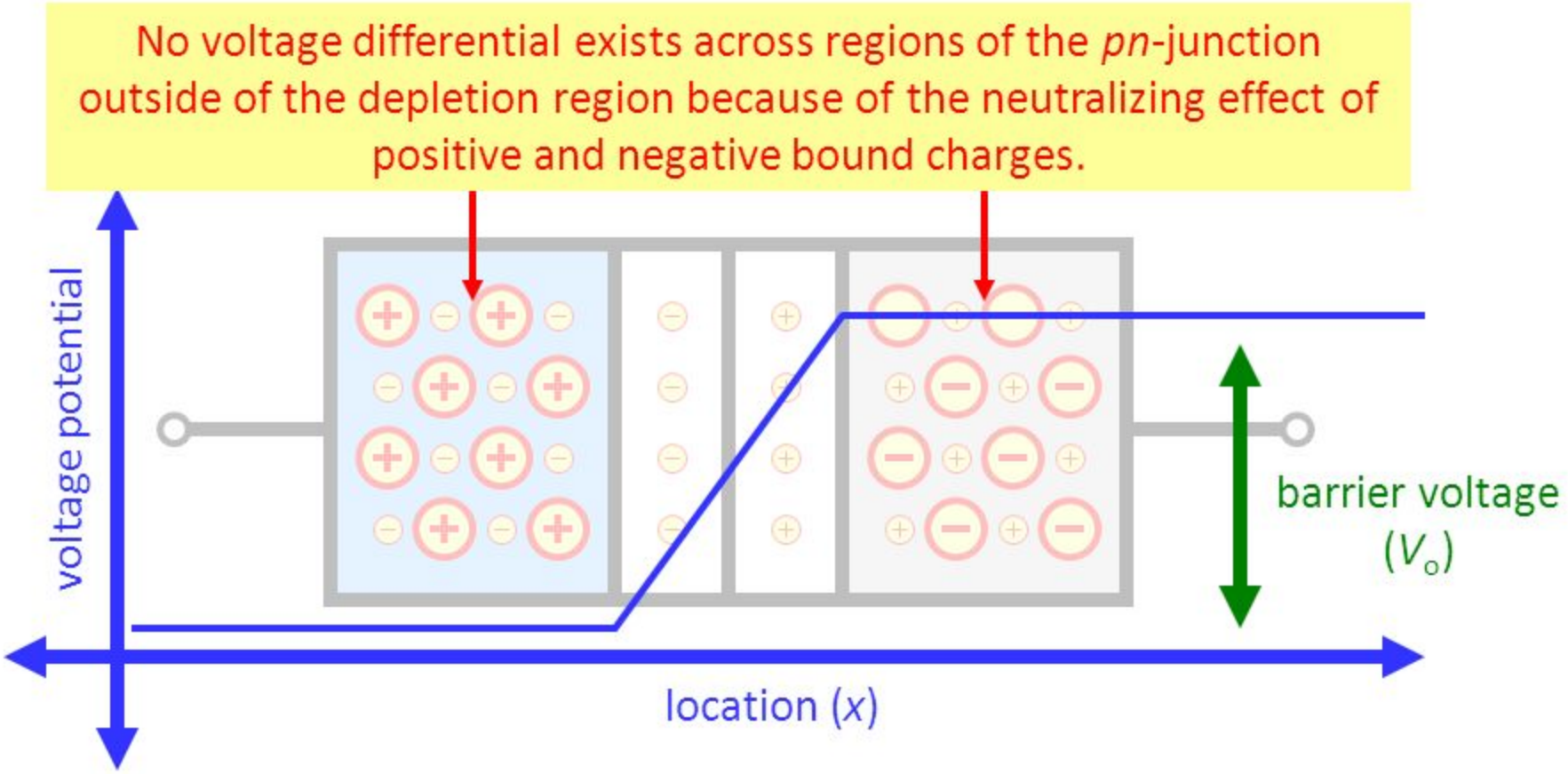


Figure: The *pn* junction with no applied voltage (open-circuited terminals).

3.4.2. **Operation with Open-Circuit Terminals**

- **Q:** Why does diffusion occur even when bound charges neutralize the electrical attraction of majority carriers to one another?
  - **A:** Diffusion current, as shown in (3.19) and (3.20), is effected by a gradient in concentration of majority carriers – not an electrical attraction of these particles to one another.

Step #4: The "uncovered" bound charges effect a voltage differential across the depletion region. The magnitude of this barrier voltage (V_0) differential grows, as diffusion continues.



Step #5: The barrier voltage (V_0) is an electric field whose polarity opposes the direction of diffusion current (I_D). As the magnitude of V_0 increases, the magnitude of I_D decreases.

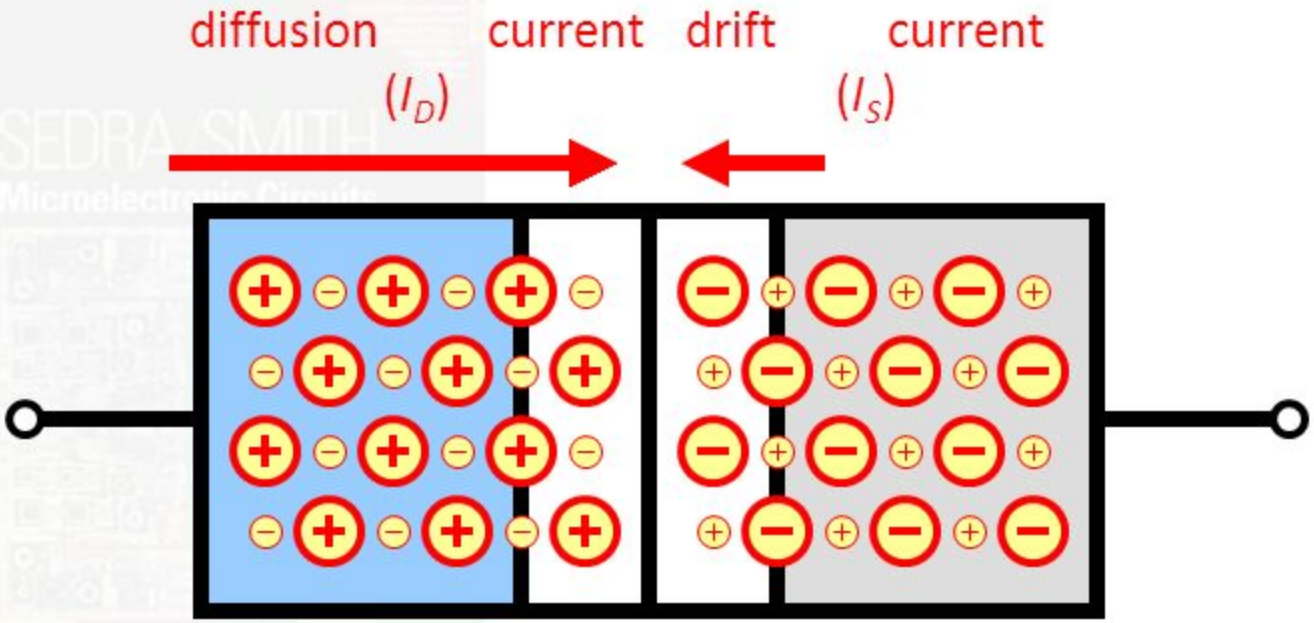
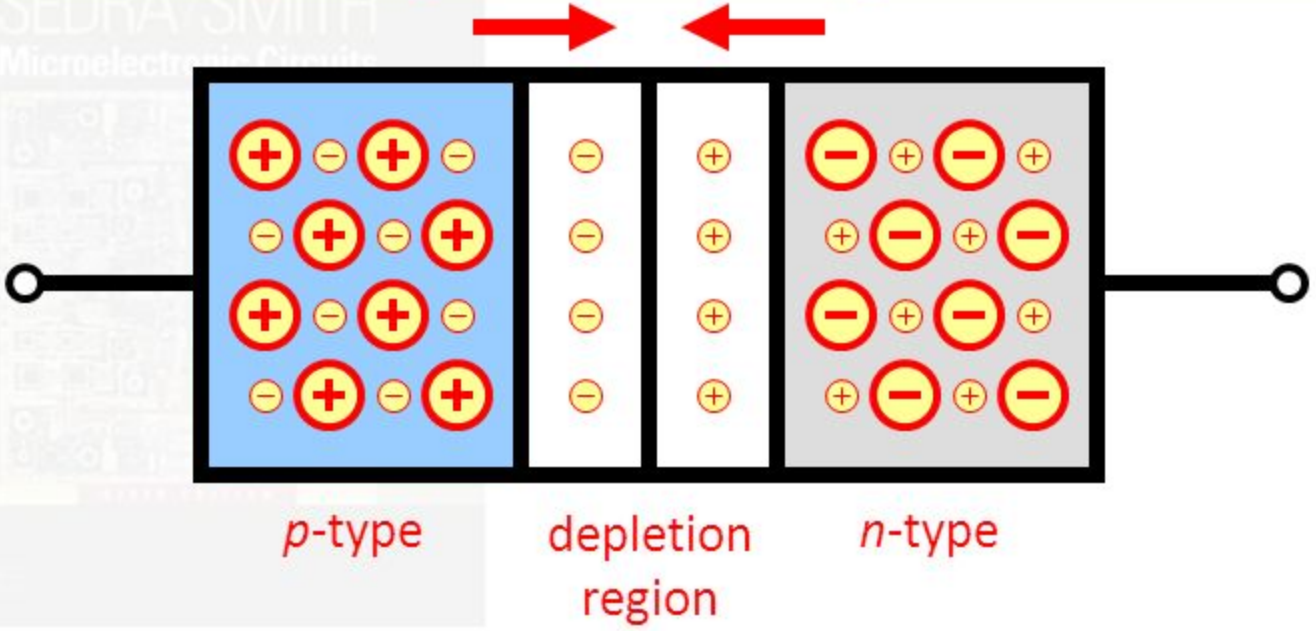


Figure: The *pn* junction with no applied voltage (open-circuited terminals).

Step #6: Equilibrium is reached, and diffusion ceases, once the magnitudes of diffusion and drift currents equal one another – resulting in no net flow.

Once equilibrium is achieved, no net current flow exists ($I_{net} = I_D - I_S$) within the *pn*-junction while under open-circuit condition.



3.4.2. Operation with Open-Circuit Terminals

- ***pn*-junction built-in voltage (V_0)** – is the equilibrium value of barrier voltage.
  - It is defined to the right.
  - Generally, it takes on a value between 0.6 and 0.9*V* for silicon at room temperature.
  - This voltage is applied across depletion region, not terminals of *pn* junction.
    - Power cannot be drawn from V_0.

V_0 = barrier voltage
V_T = thermal voltage
N_A = acceptor doping concentration
N_D = donor doping concentration
n_i = concentration of free electrons... ...in intrinsic semiconductor

(eq3.22) $$V_0 = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right)$$

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The Drift Current I_S and Equilibrium

- In addition to majority-carrier diffusion current (I_D), a component of current due to minority carrier drift exists (I_S).
- Specifically, some of the thermally generated holes in the *p*-type and *n*-type materials move toward and reach the edge of the depletion region.
- There, they experience the electric field (V_0) in the depletion region and are swept across it.
  - Unlike diffusion current, the polarity of V_0 reinforces this drift current.

3.4.2. Operation with Open-Circuit Terminals

- Because these holes are free electrons are produced by thermal energy, I_S is heavily dependent on temperature
- Any depletion-layer voltage, regardless of how small, will cause the transition across junction. Therefore I_S is independent of V_0.
- **drift current** (I_S) – is the movement of these minority carriers.
  - aka. electrons from *n*-side to *p*-side of the junction

Note that the magnitude of drift current (I_S) is unaffected by level of diffusion and / or V_0. It will be, however, affected by temperature.

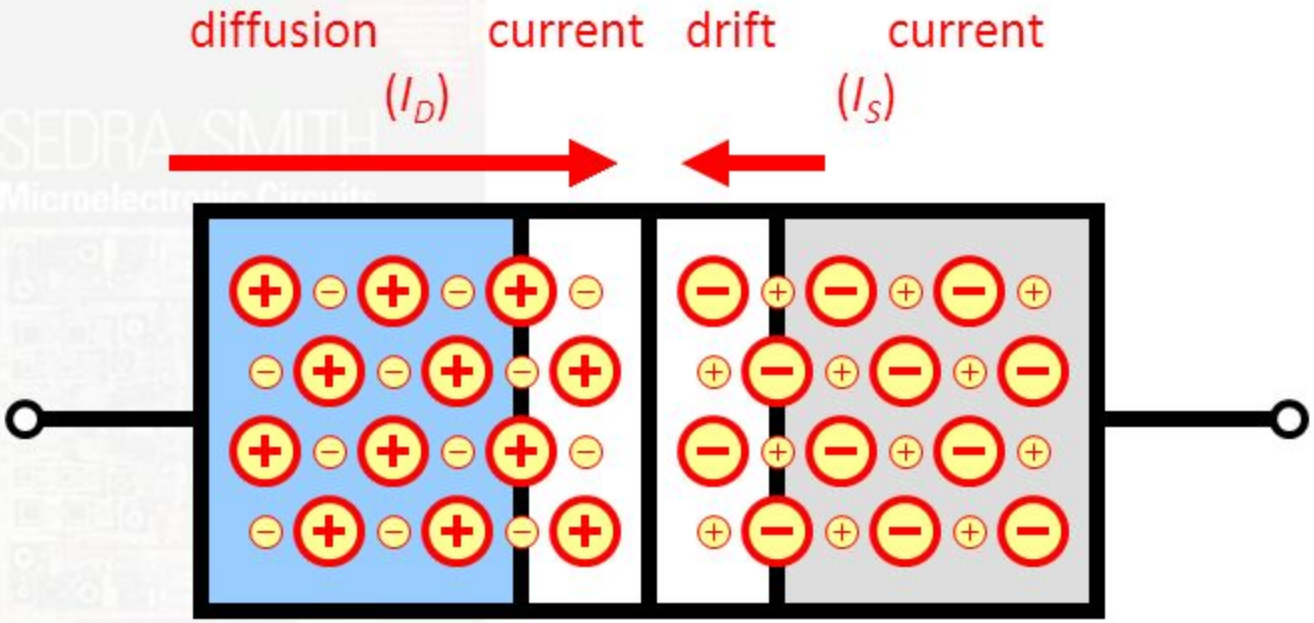


Figure: The *pn* junction with no applied voltage (open-circuited terminals).

3.4.2. Operation with Open-Circuit Terminals

- **Q:** Is the depletion region always symmetrical? As shown on previous slides?
  - **A:** The short answer is no.
- **Q:** Why?
  - Typically $N_A > N_D$
  - And, because concentration of doping agents (N_A, N_D) is unequal, the width of depletion region will differ from side to side.

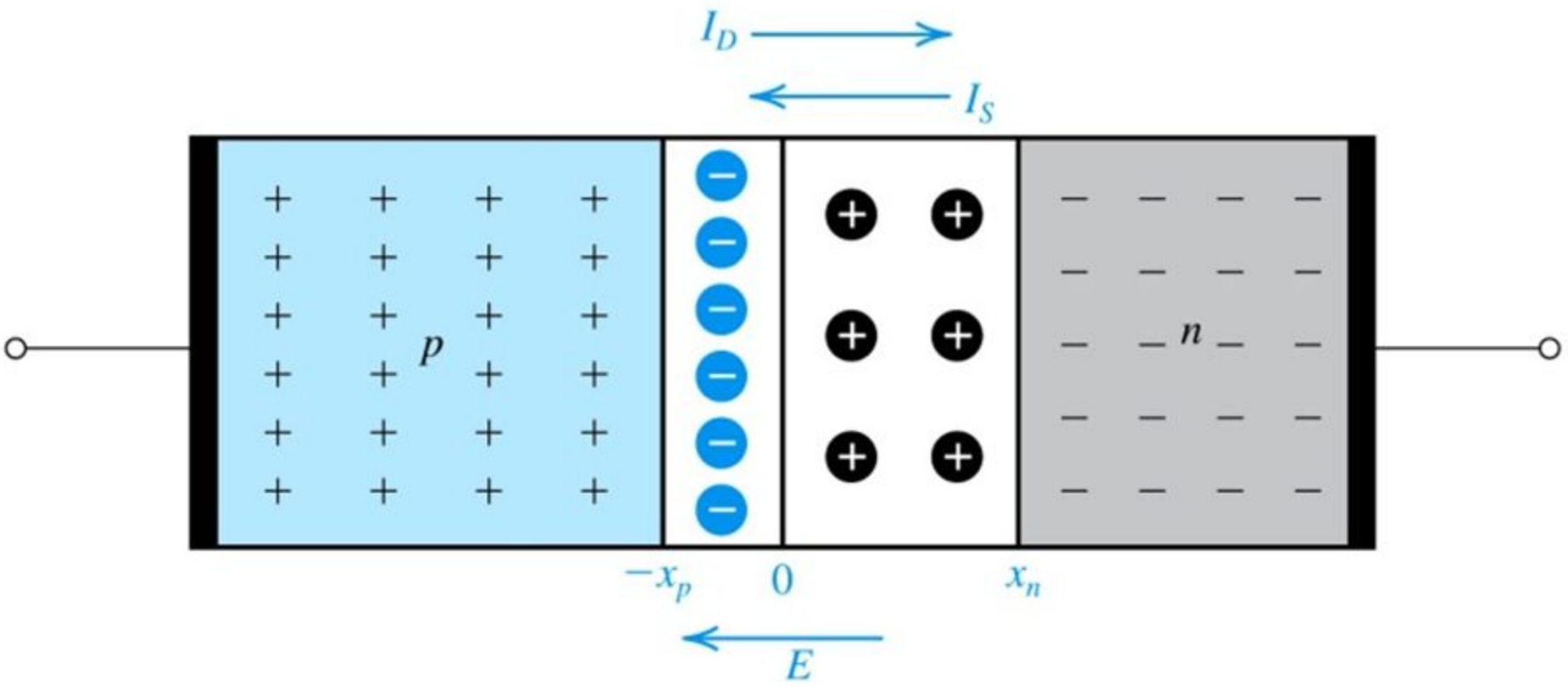
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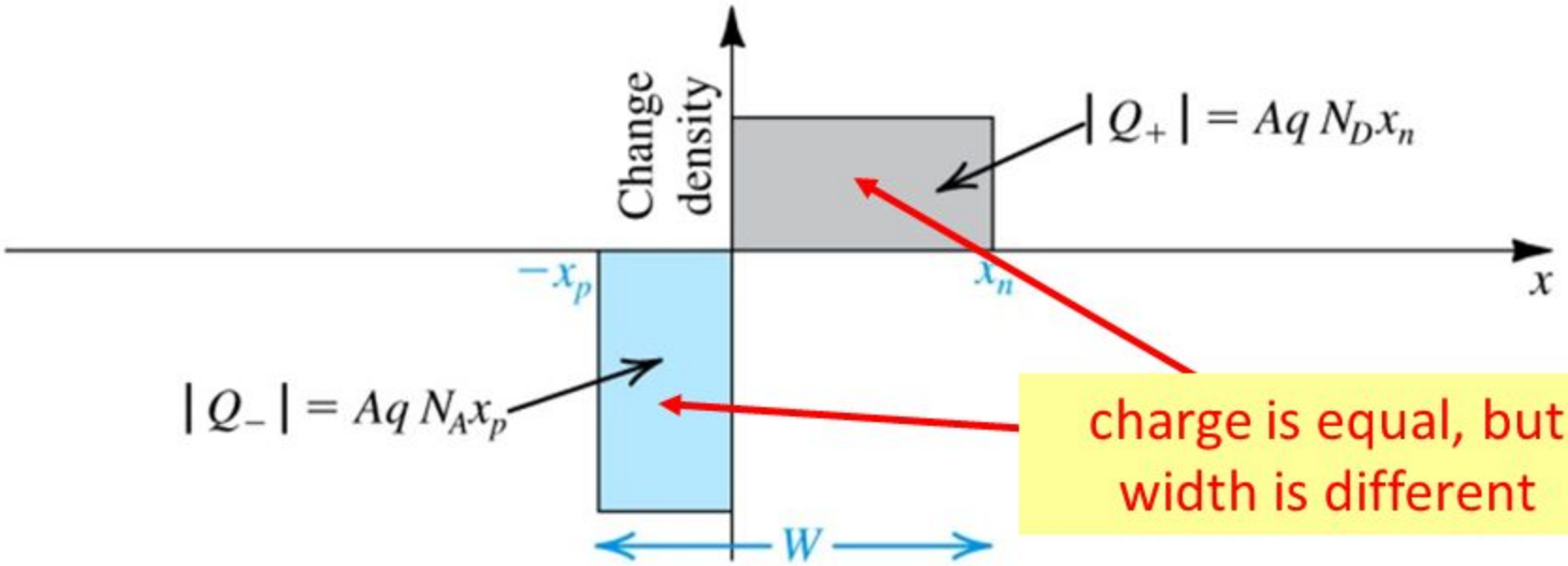
3.4.2. Operation with Open-Circuit Terminals

- **Q:** Why?
  - **A:** Because, typically $N_A > N_D$.
    - When the concentration of doping agents (N_A, N_D) is unequal, the width of depletion region will differ from side to side.
    - The depletion region will extend deeper in to the "less doped" material, a requirement to uncover the same amount of charge.
      - x_p = width of depletion *p*-region
      - x_n = width of depletion *n*-region

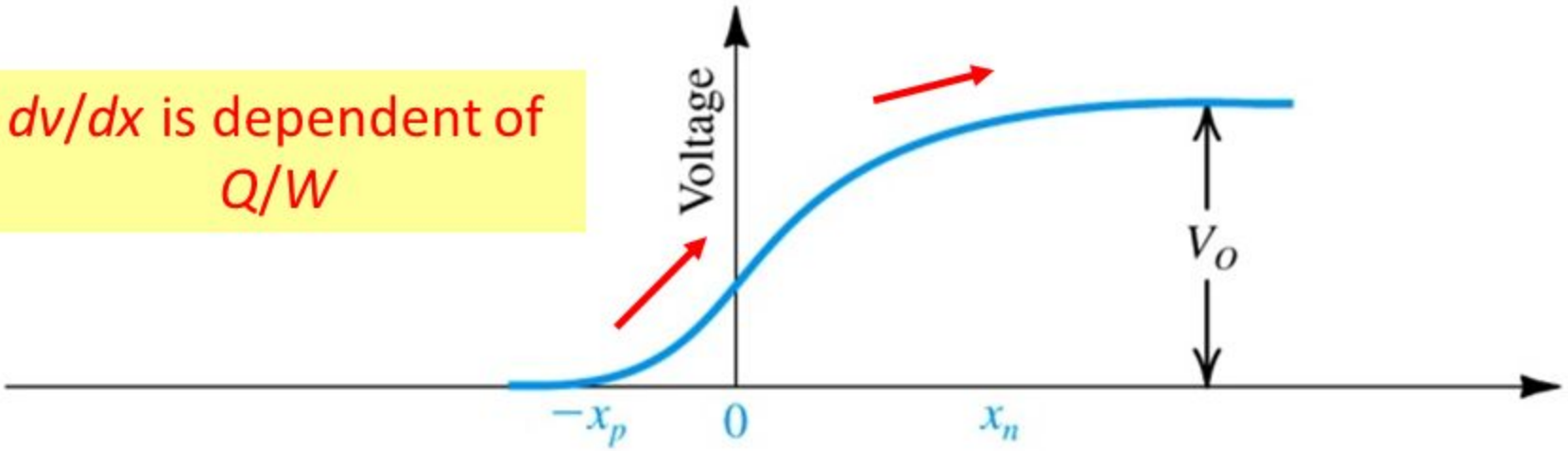
3.4.2. Operation with Open-Circuit Terminals

The depletion region will **extend further** in to region with "less" doping. However, the "number" of uncovered charges is the same.





dv/dx is dependent of *Q/W*



3.4.2. Operation with Open-Circuit Terminals

- **Q:** How is the charge stored in both sides of the depletion region defined?
  - **A:** Refer to equations to right. Note that these values should equal one another.

$|Q_+|$ = magnitude of charghe on n-side of junction

q = magnitude of electric charge

A = cross-sectional area of junction

x_n = penetration of depletion region into n-side

N_D = concentration of donor atoms

$$\text{(eq3.23)} \quad |Q_+| = qAx_nN_D$$

$$\text{(eq3.24)} \quad |Q_-| = qAx_pN_A$$

$|Q_-|$ = magnitude of charghe on n-side of junction

q = magnitude of electric charge

A = cross-sectional area of junction

x_p = penetration of depletion region into p-side

N_A = concentration of acceptor atoms

3.4.2. Operation with Open-Circuit Terminals

- **Q:** What information can be derived from this equality?
  - **A:** In reality, the depletion region exists almost entirely on one side of the *pn*-junction – due to great disparity between $N_A > N_D$.

$$qAx_pN_A = qAx_nN_D \quad \rightarrow \quad \text{(eq3.25)} \quad \frac{x_n}{x_p} = \frac{N_A}{N_D}$$

3.4.2. Operation with Open-Circuit Terminals

- Note that both x_p and x_n may be defined in terms of the depletion region width (W).

W= width of depletion region
ε_S= electrical permiability of silicon ($11.7\varepsilon_0$=1.04E–12F/cm)
q= magnitude of electron charge
N_A= concentration of acceptor atoms
N_D= concentration of donor atoms
V_0= barrier / junction built-in voltage

(eq3.26) $$W = x_n + x_p = \sqrt{\frac{2\varepsilon_S}{q}\left(\frac{1}{N_A}+\frac{1}{N_D}\right)V_0}$$

(eq3.27) $$x_n = W\frac{N_A}{N_A+N_D}$$

(eq3.28) $$x_p = W\frac{N_D}{N_A+N_D}$$

3.4.2. Operation with Open-Circuit Terminals

- Note, also, the charge on either side of the depletion region may be calculated via (3.29) and (3.30).

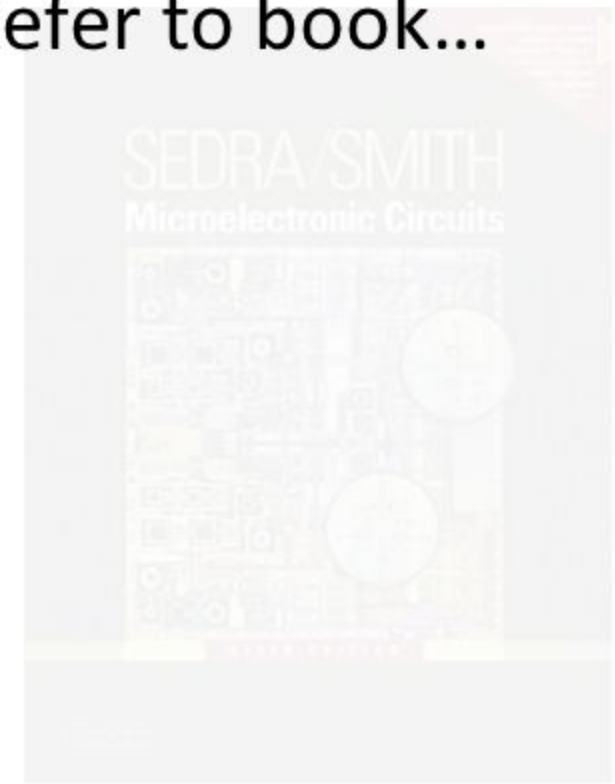
(eq3.29) $$Q_J = |Q_{\pm}| = Aq\left(\frac{N_A N_D}{N_A + N_D}\right)W$$

(eq3.30) $$Q_J = A\sqrt{2\varepsilon_S q\left(\frac{N_A N_D}{N_A + N_D}\right)V_0}$$

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Example 3.5

- Refer to book…



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3.4.2. Operation with Open-Circuit Terminals

- **Q:** What has been learned about the *pn*-junction?
  - **A:** composition
    - The *pn* junction is composed of two silicon-based semiconductors, one doped to be *p*-type and the other *n*-type.
  - **A:** majority carriers
    - Are generated by doping.
    - Holes are present on *p*-side, free electrons are present on *n*-side.

3.4.2. Operation with Open-Circuit Terminals

- **Q:** What has been learned about the *pn*-junction?
  - **A:** bound charges
    - Charge of majority carriers are neutralized electrically by bound charges.
  - **A:** diffusion current I_D
    - Those majority carriers close to the junction will diffuse across, resulting in their elimination.

3.4.2. Operation with Open-Circuit Terminals

- **Q:** What has been learned about the pn-junction?
  - **A:** depletion region
    - As these carriers disappear, they release bound charges and effect a voltage differential V_0.
  - **A:** depletion-layer voltage
    - As diffusion continues, the depletion layer voltage (V_0) grows, making diffusion more difficult and eventually bringing it to halt.

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3.4.2. Operation with Open-Circuit Terminals

- **Q:** What has been learned about the pn-junction?
  - **A:** minority carriers
    - Are generated thermally.
    - Free electrons are present on *p*-side, holes are present on *n*-side.
  - **A:** drift current I_S
    - The depletion-layer voltage (V_0) facilitates the flow of minority carriers to opposite side.
  - **A:** open circuit equilibrium $I_D = I_S$

3.5.1. Qualitative Description of Junction Operation

- Figure to right shows *pn*-junction under three conditions:
  - (a) open-circuit – where a barrier voltage V_0 exists.
  - (b) reverse bias – where a dc voltage V_R is applied.
  - (c) forward bias – where a dc voltage V_F is applied.

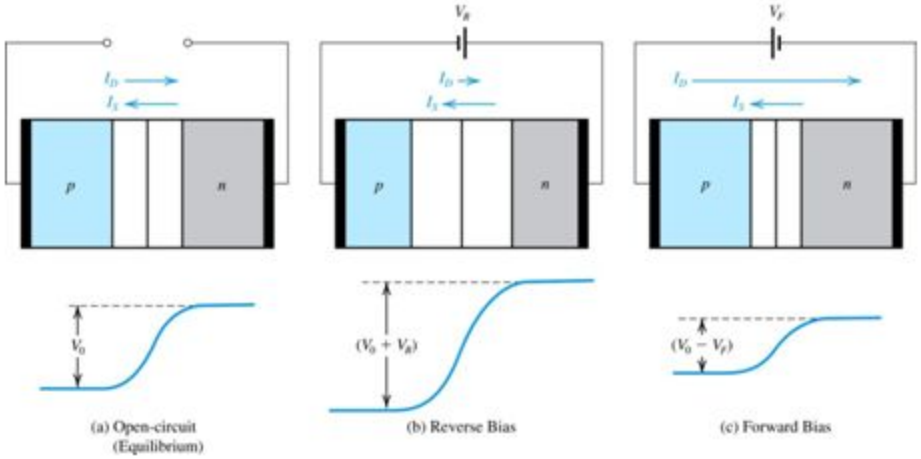


Figure 3.11: The *pn* junction in: **(a)** equilibrium; **(b)** reverse bias; **(c)** forward bias.

1) no voltage applied

2) voltage differential across depletion zone is V_0

3) $I_D = I_S$

1) negative voltage applied

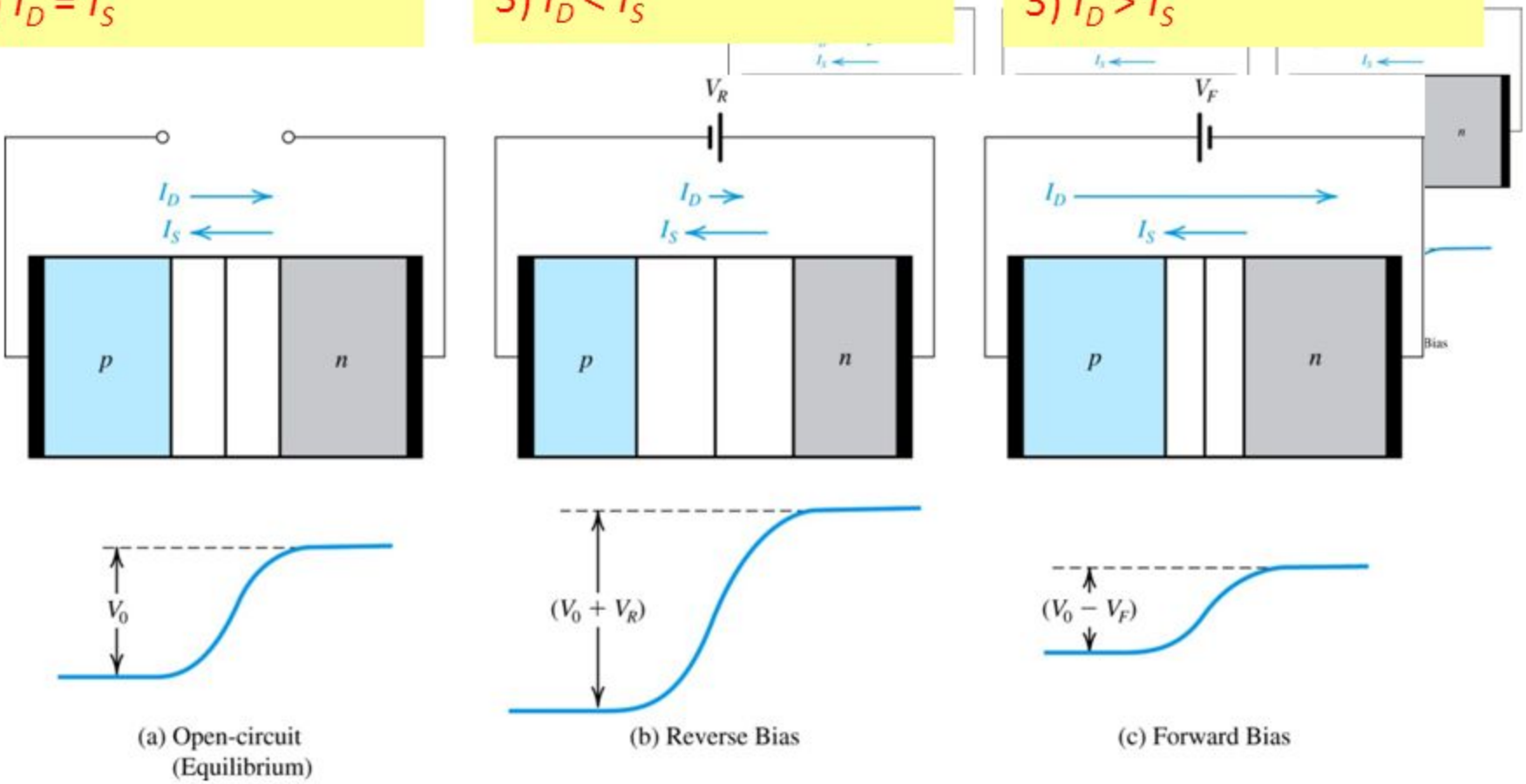
2) voltage differential across depletion zone is $V_0 + V_R$

3) $I_D < I_S$

1) positive voltage applied

2) voltage differential across depletion zone is $V_0 - V_F$

3) $I_D > I_S$



(a) Open-circuit (Equilibrium)

(b) Reverse Bias

(c) Forward Bias

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3.5.1. Qualitative Description of Junction Operation

- **reverse bias case**
  - the externally applied voltage V_R adds to (aka. reinforces) the barrier voltage V_0
    - …increase effective barrier
  - this reduces rate of diffusion, reducing I_D
    - if $V_R > 1V$, I_D will fall to $0A$
  - the drift current I_S is unaffected, but dependent on temperature

minimal current flows in reverse-bias case

- **forward bias case**
  - the externally applied voltage V_F subtracts from the barrier voltage V_0
    - …decrease effective barrier
  - this increases rate of diffusion, increasing I_D
  - the drift current I_S is unaffected, but dependent on temperature

significant current flows in forward-bias case

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Forward-Bias Case

- Observe that decreased barrier voltage will be accompanied by…
  - (1) decrease in stored uncovered charge on both sides of junction
  - (2) smaller depletion region
- Width of depletion region shown to right.

W = width of depletion region

ε_S = electrical permiability of silicon ($11.7\varepsilon_0 = 1.04\mathbf{E}{-}12\,F/cm$)

q = magnitude of electron charge

N_A = concentration of acceptor atoms

N_D = concentration of donor atoms

V_0 = barrier / junction built-in voltage

V_F = externally applied forward-bias voltage

$$W = x_n + x_p = \sqrt{\frac{2\varepsilon_S}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)\underbrace{(V_0 - V_F)}_{\substack{\text{action:}\\ \text{replace } V_0\\ \text{with } V_0 - V_F}}}$$

$$Q_J = A\sqrt{2\varepsilon_S q\left(\frac{N_A N_D}{N_A + N_D}\right)\underbrace{(V_0 - V_F)}_{\substack{\text{action:}\\ \text{replace } V_0\\ \text{with } V_0 - V_F}}}$$

Q_J = magnitude of charge stored on either side of depletion region

Reverse-Bias Case

- Observe that increased barrier voltage will be accompanied by…
  - (1) increase in stored uncovered charge on both sides of junction
  - (2) wider depletion region
- Width of depletion region shown to right.

W = width of depletion region
ε_S = electrical permiability of silicon ($11.7\varepsilon_0 = 1.04\mathrm{E}{-12}\,F/cm$)
q = magnitude of electron charge
N_A = concentration of acceptor atoms
N_D = concentration of donor atoms
V_0 = barrier / junction built-in voltage
V_R = externally applied reverse-bias voltage

$$\text{(eq3.31)}\quad W = x_n + x_p = \sqrt{\frac{2\varepsilon_S}{q}\left(\frac{1}{N_A}+\frac{1}{N_D}\right)\underbrace{(V_0+V_R)}_{\substack{\text{action:}\\ \text{replace } V_0\\ \text{with } V_0+V_R}}}$$

$$\text{(eq3.32)}\quad Q_j = A\sqrt{2\varepsilon_S q\left(\frac{N_A N_D}{N_A+N_D}\right)\underbrace{(V_0+V_R)}_{\substack{\text{action:}\\ \text{replace } V_0\\ \text{with } V_0+V_R}}}$$

Q_j = magnitude of charge stored on either side of depletion region

3.5.2. The Current-Voltage Relationship of the Junction

- **Q:** What happens, exactly, when a forward-bias voltage (V_F) is applied to the *pn*-junction?
  - **step #1:** Initially, a small forward-bias voltage (V_F) is applied. It, because of its polarity, pushes majority carriers (holes in *p*-region and electrons in *n*-region) toward the junction and reduces width of the depletion zone.
    - Note, however, that this force is opposed by the built-in voltage built in voltage V_0.

step #1: Initially, a small forward-bias voltage (V_F) is applied. It,

Note that, in this figure, the smaller circles represent minority carriers and **not bound charges** – which are not considered here.

the depletion zone.

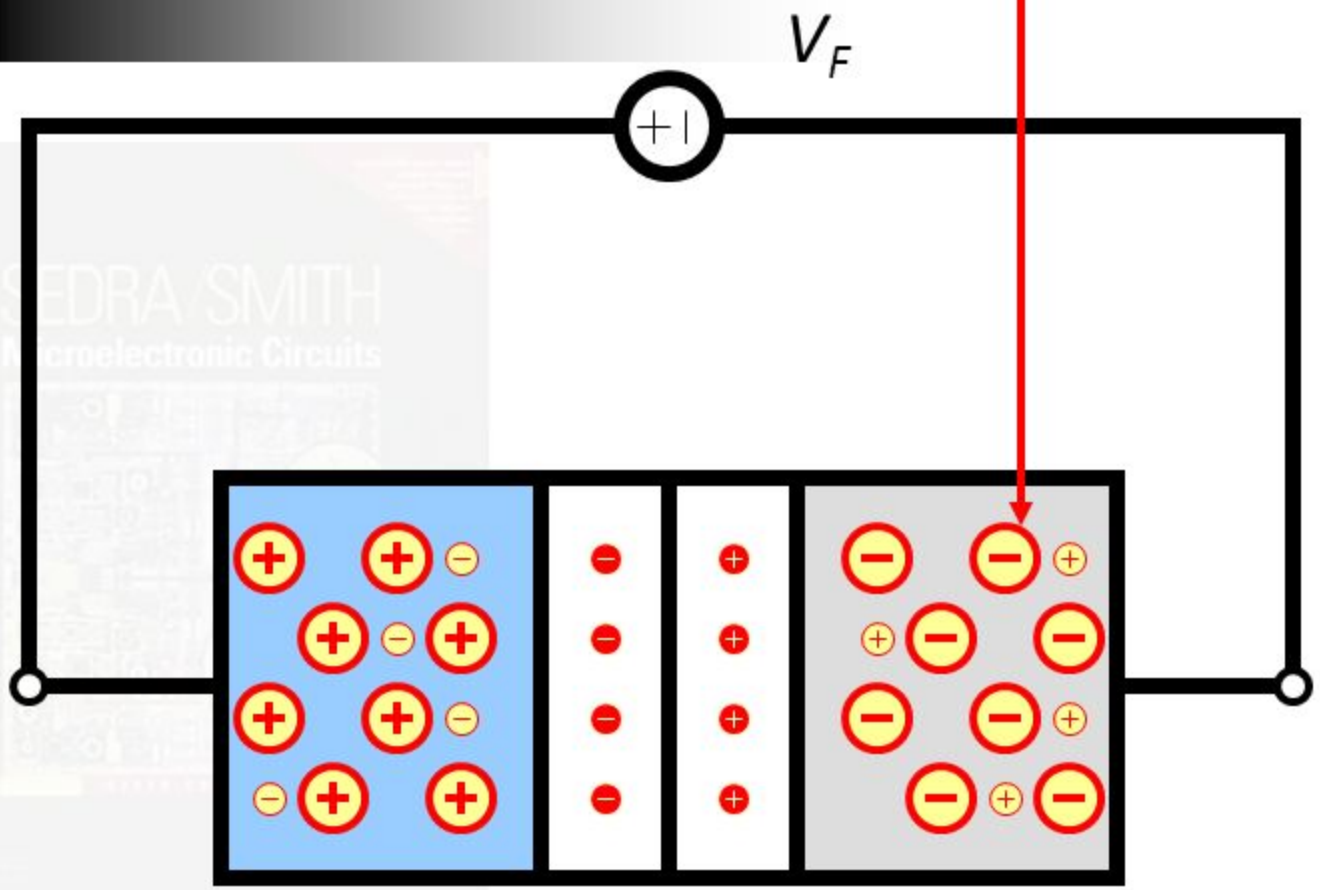
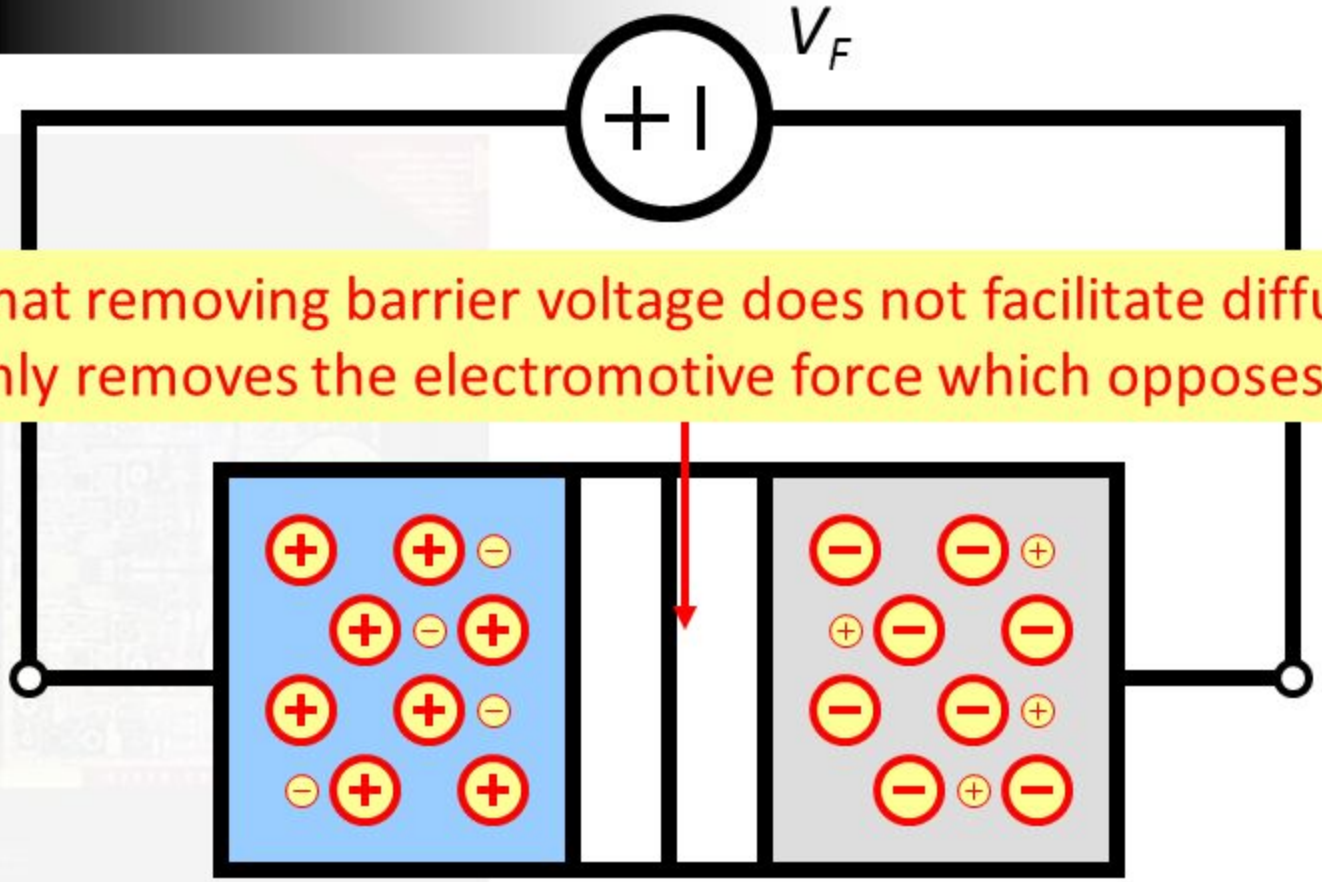


Figure: The *pn* junction with applied voltage.

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step #2: As the magnitude of V_F increases, the depletion zone becomes thin enough such that the barrier voltage ($V_0 - V_F$) cannot stop diffusion current – as described in previous slides.



Note that removing barrier voltage does not facilitate diffusion, it only removes the electromotive force which opposes it.

Figure: The *pn* junction with applied voltage.

step #3: Majority carriers (free electrons in *n*-region and holes in *p*-region) cross the junction and become minority charge carriers in the near-neutral region.

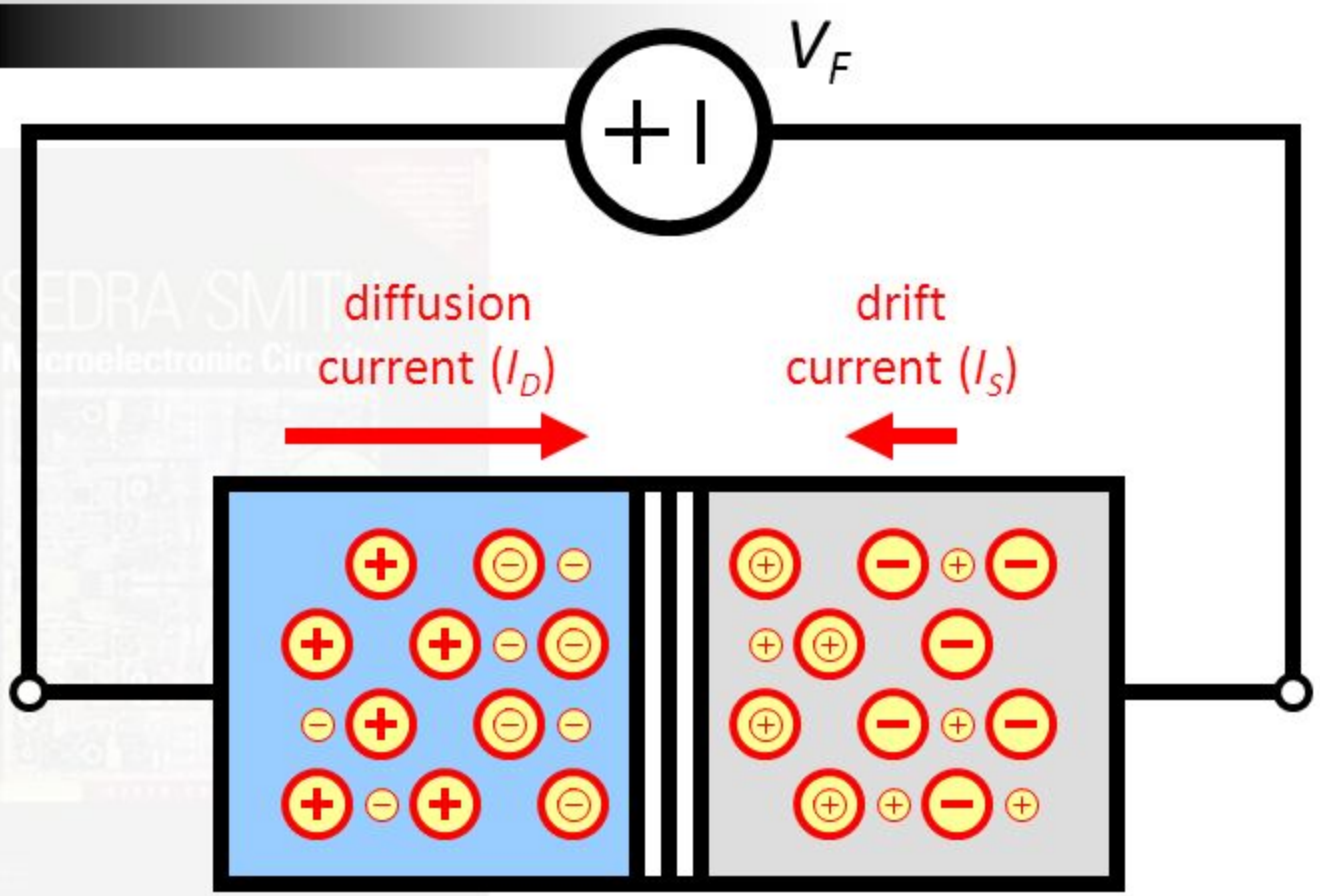


Figure: The *pn* junction with applied voltage.

step #4: The concentration of minority charge carriers increases on either side of the junction. A steady-state

For the open-circuit condition, minority carriers are evenly distributed throughout the non-depletion regions. This concentration is defined as either n_{p0} or p_{n0}.

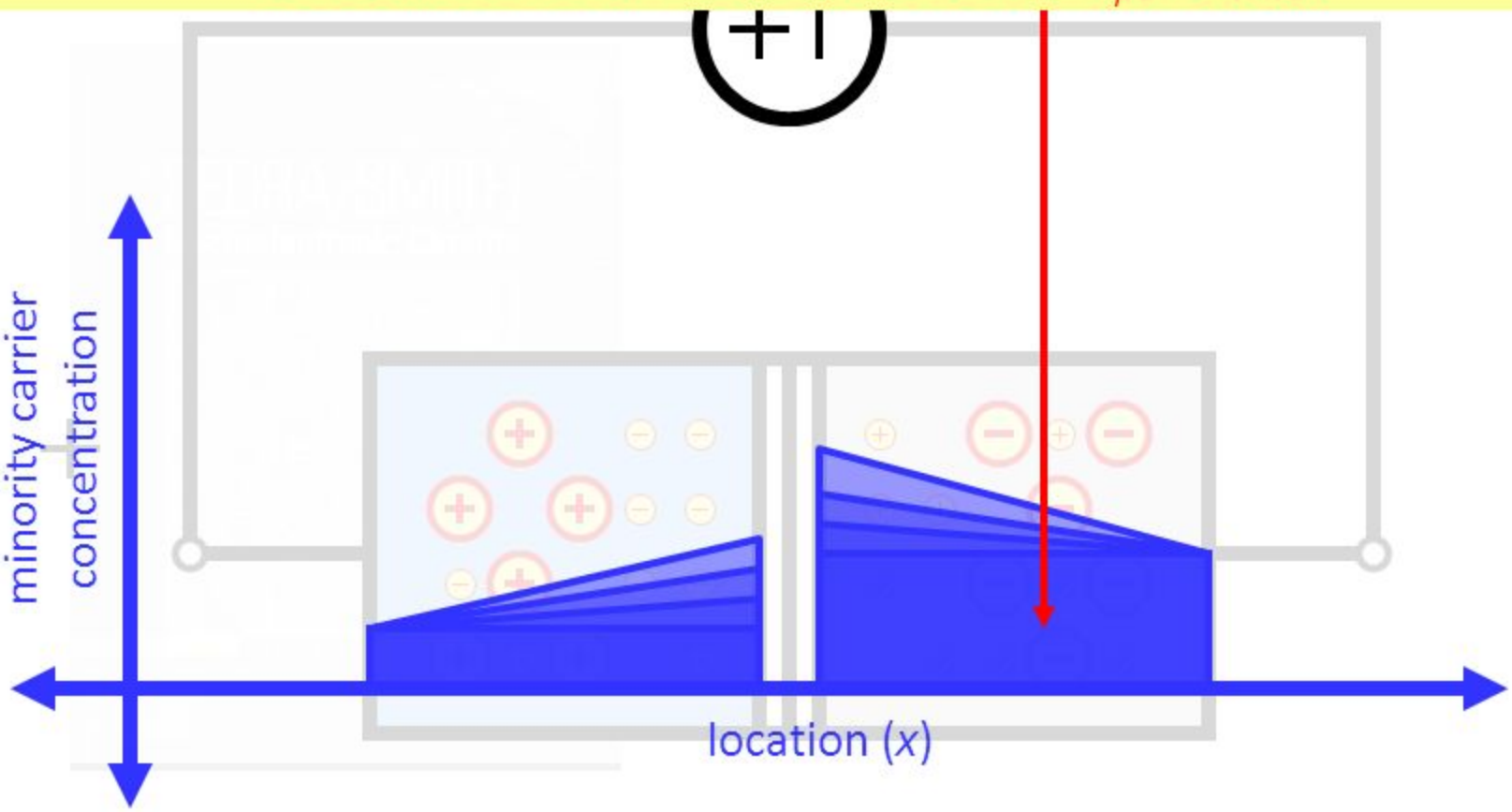


Figure: The *pn* junction with applied voltage.

step #4: The concentration of minority charge carriers increases on either side of the junction. A steady-state gradient is reached as rate of majority carriers crossing the junction equals that of recombination.

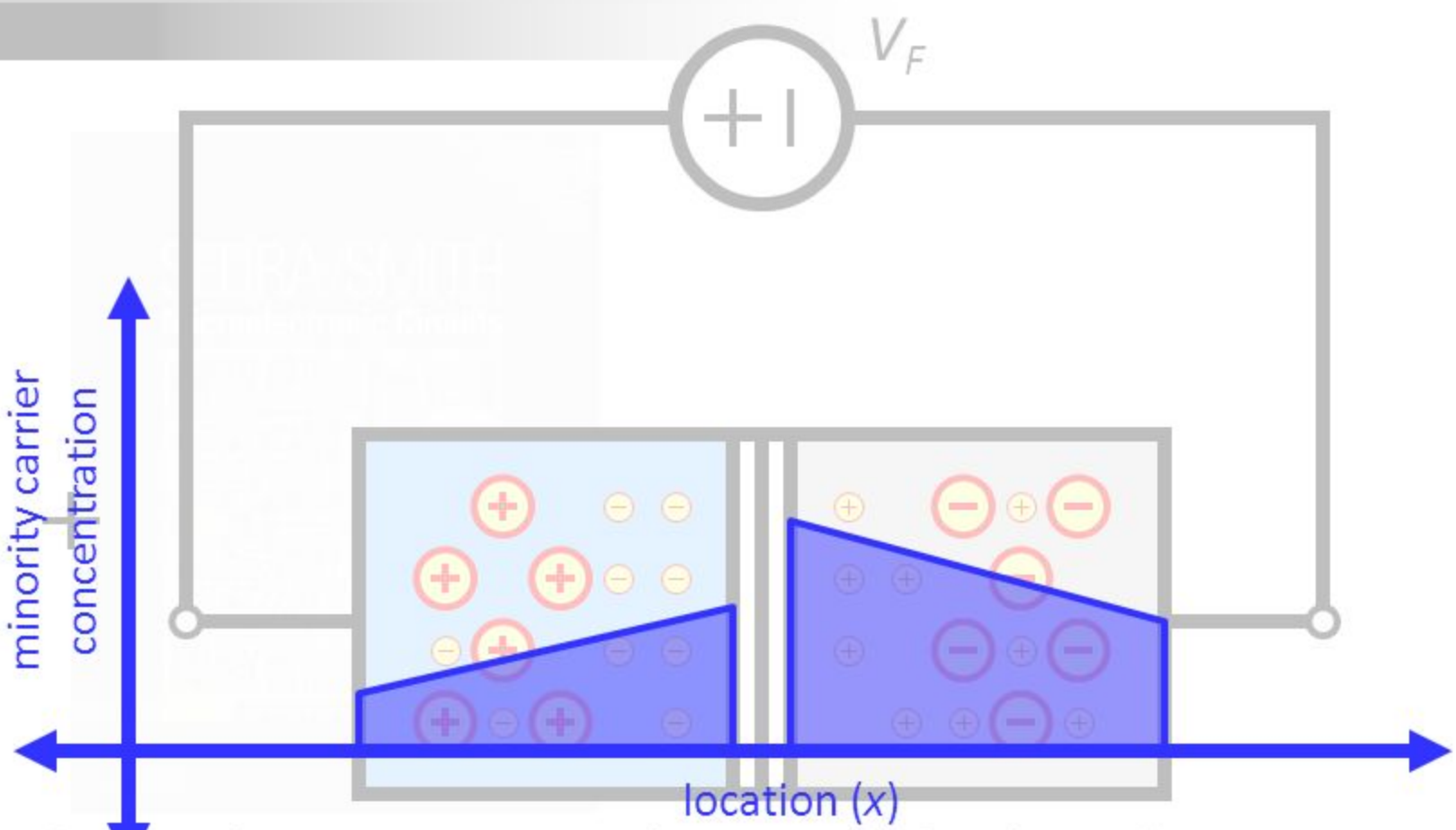


Figure: The *pn* junction with no applied voltage (open-circuited terminals).

step #5+: Diffusion current is maintained – in spite low diffusion lengths (e.g. microns) and recombination – by constant flow of both free electrons and holes towards the junction.

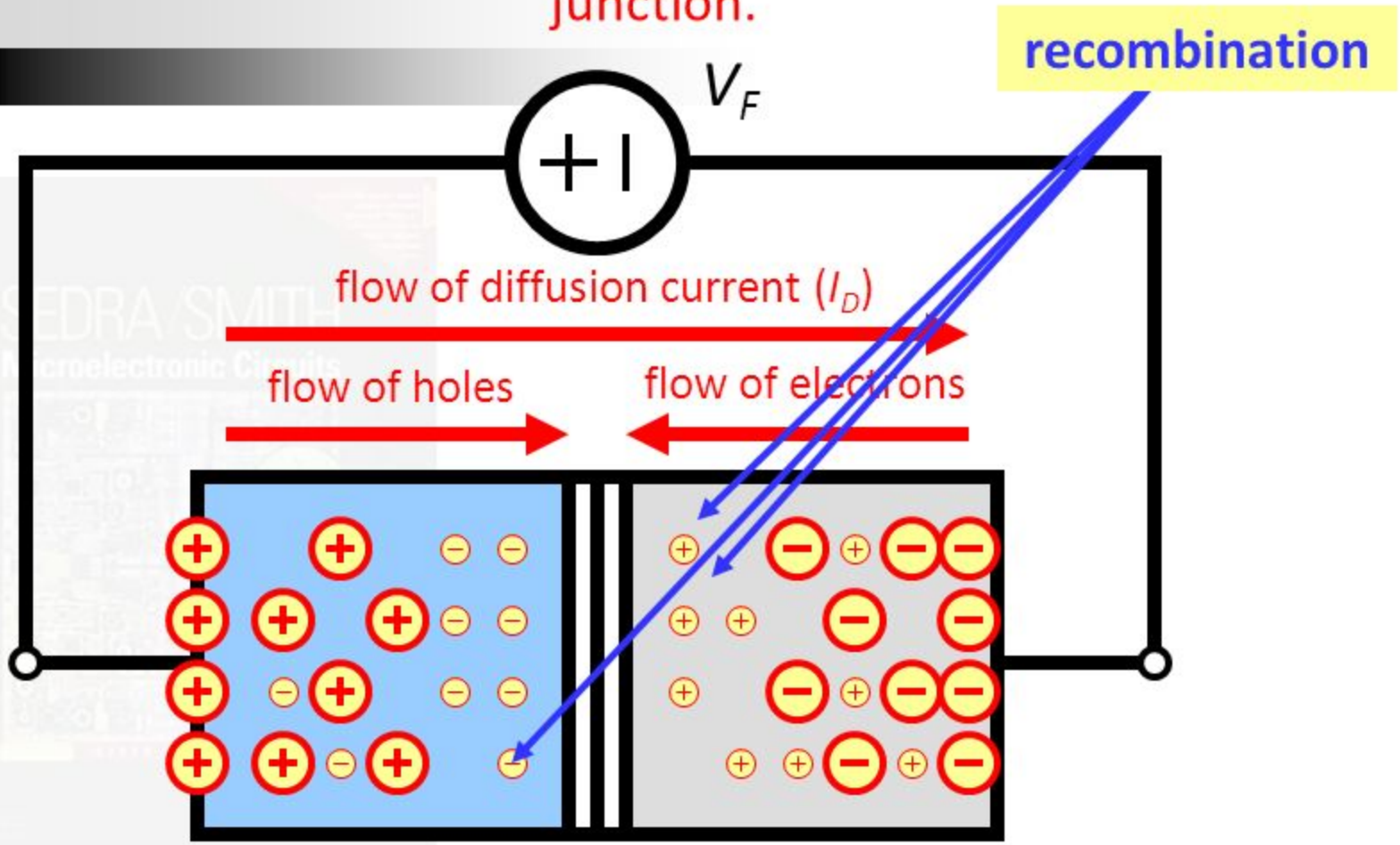


Figure: The *pn* junction with no applied voltage (open-circuited terminals).

The key aspect of (3.33) is that it relates the minority-charge carrier concentration at the junction boundary in terms of majority-charge carrier on the opposite side.

$$(\text{eq3.7})\quad p_{n0} = \frac{n_i^2}{N_A}$$

- **Q:** How is the relationship between forward-bias voltage applied (V) and minority-carrier holes and electrons defined?
  - **step #1:** Employ (3.33).
    - This function describes maximum minority carrier concentration at junction.
  - **step #2:** Subtract p_{n0} from $\boldsymbol{p}_n(x)$ to calculate the excess minority charge carriers.

$\mathbf{p}_n(x_n)$ = concentration of holes in n-region as function of x_n
p_{n0} = thermal equilibrium concentration
V = applied foward-bias voltage
V_T = thermal voltage

$$(\text{eq3.33})\quad \mathbf{p}_n(x_n) = p_{n0}e^{V/V_T}$$

$$(\text{eq3.34})\quad \text{excess concentration} = p_{n0}e^{V/VT} - p_{n0}$$

$$\text{excess concentration} = p_{n0}(e^{V/VT} - 1)$$

3.5.2. The Current-Voltage Relationship of the Junction

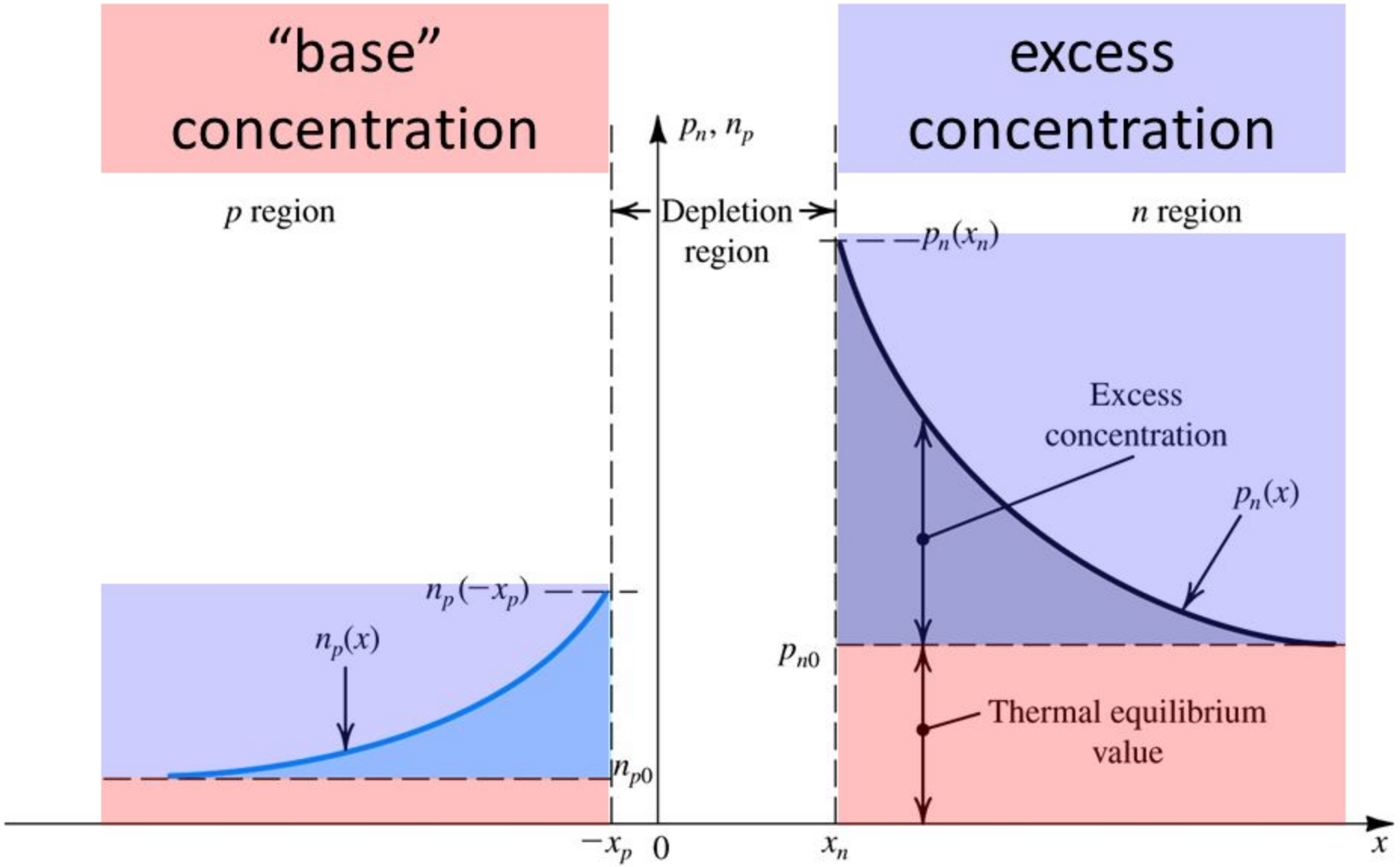
- **Q:** How is the relationship between forward-bias voltage applied (V) and minority-carrier holes and electrons defined?
  - **step #3:** Refer to (3.35).
    - This function describes the minority carrier concentration as a function of location (x), boundary of depletion region (x_n), and **diffusion length** (L_p).

$\mathbf{p}_n(x_n)$ = concentration of holes in n-region as function of x_n, p_{n0} = thermal equilibrium concentration
x = point of interest, x_n = edge of depletion region, L_p = diffusion length

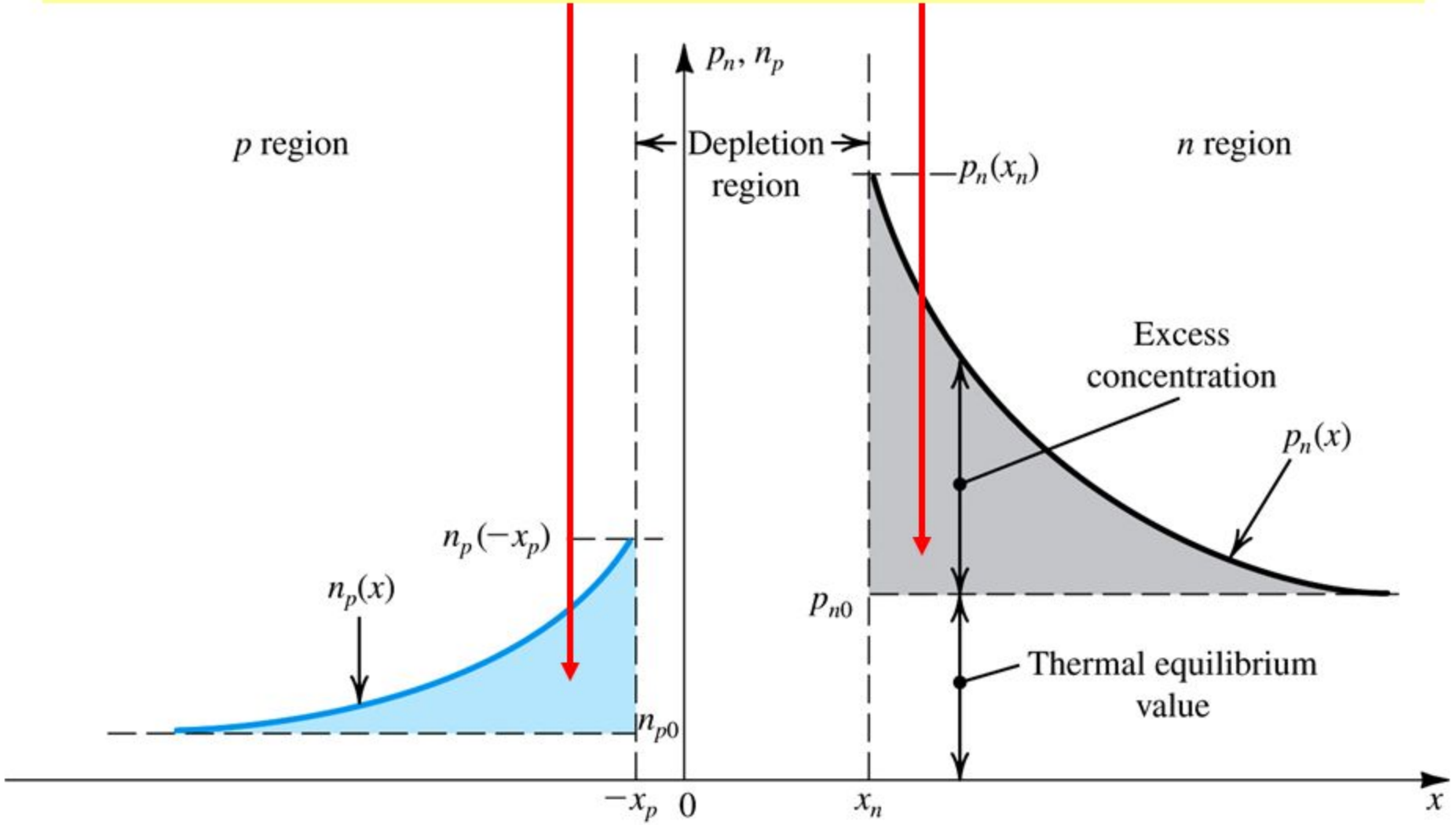
$$\text{(eq3.35)}\quad \mathbf{p}_n(x_n) = p_{n0} + \underbrace{(excess\ concentration)}_{p_{n0}(e^{V/VT}-1)} e^{-(x-x_n)/L_p}$$

$$\text{(eq3.35)}\quad \mathbf{p}_n(x_n) = p_{n0} + p_{n0}(e^{V/VT}-1)e^{-(x-x_n)/L_p}$$

steady-state minority carrier concentration on both sides of a *pn*-junction for which $N_A >> N_D$



These excess concentrations effect steady-state diffusion current. **However, how is this diffusion current defined?**



3.5.2. The Current-Voltage Relationship of the Junction

- **Q:** For forward-biased case, how is diffusion current (I_D) defined?
  - **step #1:** Take derivative of (3.35) to define component of diffusion current attributed to flow of holes.
  - **step #2:** Note that this value is maximum at $x = x_n$.

action: take derivative of $p_n(x)$

$$\frac{d\mathbf{p}_n(x)}{dx} = \underbrace{\frac{d}{dx}[p_{n0}]}_{0} + \ldots$$

$$\ldots + \underbrace{\frac{d}{dx}\left[p_{n0}(e^{V/V_T}-1)e^{-(x-x_n)/L_p}\right]}_{-\frac{p_{n0}}{L_p}(e^{V/VT}-1)e^{-(x-x_n)/L_p}}$$

action: substitute in value from above

(eq3.36) $$J_p = -qD_p\underbrace{\left(-\frac{p_{n0}}{L_p}(e^{V/V_T}-1)e^{-(x-x_n)/L_p}\right)}_{\frac{d\mathbf{p}_n(x)}{dx}}$$

action: calculate maximum

$$\mathbf{max}(J_p) = q\frac{D_p}{L_p}p_{n0}(e^{V/V_T}-1)$$

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Q: For forward-biased case, how is diffusion current defined?

- **step #3:** Define the component of maximum diffusion current attributed to minority-carrier electrons – in method similar above.

(eq3.37) maximum hole - diffusion concentration:

$$\mathbf{J}_p(+x_n) = q\frac{D_p}{L_p}p_{n0}(e^{V/V_T} - 1)$$

(eq3.38) maximum electron - diffusion concentration:

$$\mathbf{J}_n(-x_p) = q\frac{D_n}{L_n}n_{p0}(e^{V/V_T} - 1)$$

Q: For forward-biased case, how is diffusion current defined?

total current (I) through junction is equal to area (A) times maximum hole (J_p) and electron-diffusion (J_n) current densities

$$I = A\left[\mathbf{J}_p(+x_n) + \mathbf{J}_n(-x_p)\right]$$

$$I = A\left[q\frac{D_p}{L_p}p_{n0} + q\frac{D_n}{L_n}n_{p0}\right](e^{V/V_T} - 1)$$

action: substitute in values for $J_p(+x_n)$ and $J_n(-x_p)$

$$I = Aqn_i^2\left[\frac{D_p}{L_pN_D} + \frac{D_n}{L_nN_A}\right](e^{V/V_T} - 1)$$

action: substitute $p_{n0} = n_i^2/N_D$ and $n_{p0} = n_i^2/N_A$

$$I = I_S(e^{V/V_T} - 1)$$

action: substitute $I_s = Aqn_i^2\left[\frac{D_p}{L_pN_D} + \frac{D_n}{L_nN_A}\right]$

- **step #4:** Define total diffusion current as sum of components attributed to free electrons and holes.

3.5.2. The Current-Voltage Relationship of the Junction

- **Q:** For forward-biased case, how is diffusion current (I_D) defined?
  - **A:** Refer to (3.40). This is an important equation which will be employed in future chapters.

$$\text{(eq3.40)}\quad I = \underbrace{\left(Aqn_i^2 \left[\frac{D_p}{L_p N_D} + \frac{D_n}{L_n N_A} \right] \right)}_{I_S} \left(e^{V/V_T} - 1 \right) = I_S \left(e^{V/V_T} - 1 \right)$$

3.5.2. The Current-Voltage Relationship of the Junction

- **Q:** Why is diffusion current (I_D) dependent on the concentration gradient of minority (as opposed to majority) charge carriers?
  - **A:** Essentially, it isn't.
    - Equation (3.33) defines the minority-charge carrier concentration in terms of the majority-charge carrier concentrations in "other" region.
    - As such, the diffusion current (I_D) is most dependent on two factors: applied forward-bias voltage (V_F) and doping.

3.5.2. The Current-Voltage Relationship of the Junction

(eq3.40) $I = I_S(e^{V/V_T} - 1)$

- **saturation current** (I_S) – is the maximum reverse current which will flow through *pn*-junction.
  - It is proportional to cross-section of junction (A).
  - Typical value is $10^{-18} A$.

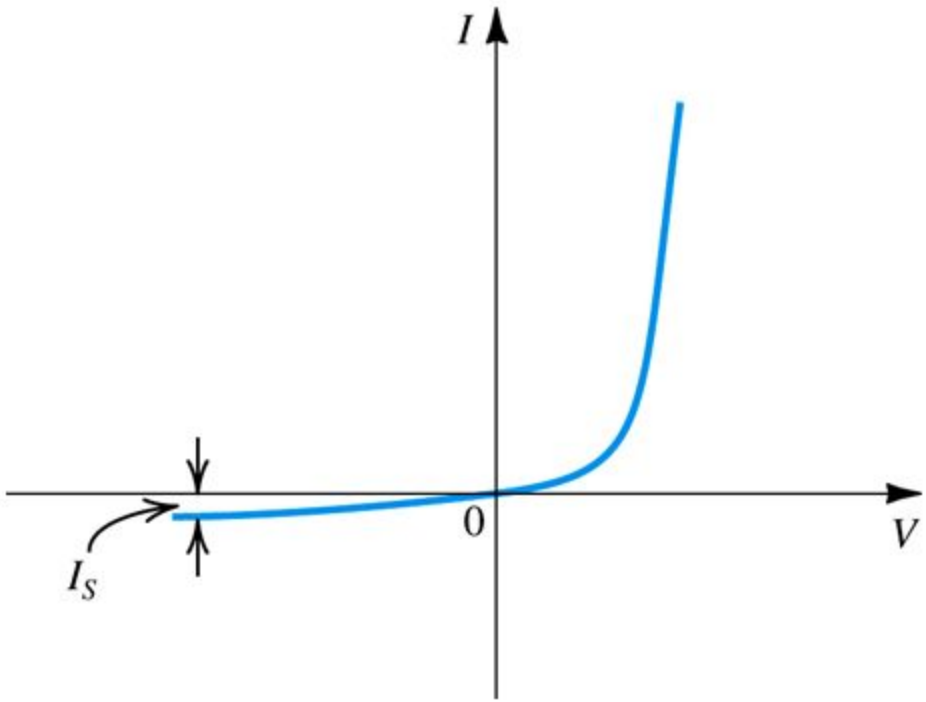


Figure 3.13: The *pn* junction *I*–*V* characteristic.

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Example 3.6: *pn*-Junction

- Consider a forward-biased *pn* junction conducting a current of $I = 0.1mA$ with following parameters:
  - $N_A = 10^{18}/cm^3$, $N_D = 10^{16}/cm^3$, A = $10^{-4}cm^2$, n_i = 1.5**E**10$/cm^3$, $L_p = 5um$, $L_n = 10um$, D_p (*n*-region) = $10cm^2/s$, D_n (*p*-region) = $18cm^2/s$
- **Q(a):** Calculate I_S.
- **Q(b):** Calculate the forward bias voltage (V).
- **Q(c):** Component of current I due to hole injection and electron injection across the junction

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Summary (1)

- Today's microelectronics technology is almost entirely based on the semiconductor silicon. If a circuit is to be fabricated as a monolithic integrated circuit (IC), it is made using a single silicon crystal, no matter how large the circuit is.
- In a crystal of intrinsic or pure silicon, the atoms are held in position by covalent bonds. At very low temperatures, all the bonds are intact; No charge carriers are available to conduct current. As such, at these low temperatures, silicone acts as an insulator.

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Summary (2)

- At room temperature, thermal energy causes some of the covalent bonds to break, thus generating free electrons and holes that become available to conduct electricity.
- Current in semiconductors is carried by free electrons and holes. Their numbers are equal and relatively small in intrinsic silicon.
- The conductivity of silicon may be increased drastically by introducing small amounts of appropriate impurity materials into the silicon crystal – via process called doping.

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Summary (3)

- There are two kinds of doped semiconductor: *n*-type in which electrons are abundant, *p*-type in which holes are abundant.
- There are two mechanisms for the transport of charge carriers in a semiconductor: drift and diffusion.
- Carrier drift results when an electric field (*E*) is applied across a piece of silicon. The electric field accelerates the holes in the direction of *E* and electrons oppositely. These two currents sum to produce drift current in the direction of *E*.

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Summary (4)

- Carrier diffusion occurs when the concentration of charge carriers is made higher in one part of a silicon crystal than others. To establish a steady-state diffusion current, a carrier concentration must be maintained in the silicon crystal.
- A basic semiconductor structure is the *pn*-junction. It is fabricated in a silicon crystal by creating a *p*-region in proximity to an *n*-region. The *pn*-junction is a diode and plays a dominant role in the structure and operation of transistors.

Summary (5)

- When the terminals of the *pn*-junction are left open, no current flows externally. However, two equal and opposite currents (I_D and I_S) flow across the junction. Equilibrium is maintained by a built-in voltage (V_0). Note, however, that the voltage across an open junction is 0*V*, since V_0 is cancelled by potentials appearing at the metal-to-semiconductor connection interfaces.
- The voltage V_0 appears across the depletion region, which extends on both sides of the junction.

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Summary (6)

- The drift current I_S is carried by thermally generated minority electrons in the *p*-material that are swept across the depletion region into the n-side. The opposite occurs in the *n*-material. I_S flows from *n* to *p*, in the reverse direction of the junction. Its value is a strong function of temperature, but independent of V_0.
- Forward biasing of the *pn*-junction, that is applying an external voltage that makes *p* more positive than *n*, reduces the barrier voltage to $V_0 - V$ and results in an exponential increase in I_D (while I_S remains unchanged).